MODULE 8 TOYOTA ANALYSIS

Toyota

(includes Lexus and Scion)

- Toyota began phasing in EDR's as early as 2000. The first model with EDR, the Lexus GS, had both Delta V and pre-crash data.
- Some 2001-2010 models have only longitudinal Delta V. By 2011 ALL have pre-crash data.
- All Toyotas can record rear impacts.
- Gen 1 & 2 recording threshold is wake-up.
- In addition to regular interval precrash data, Toyotas record an ADDITIONAL data point at or near algorithm enable, which Toyota calls TRG (trigger).

Toyota

(includes Lexus and Scion)

- Records frontal data in frontal algo wakeups/events, side data in side wakeups/events if equipped, from their respective AE's. Front & Side NOT on same time zero.
- Side data has shorter time than frontal data until 13EDR
- Rollover events if equipped, from rollover algo AE.
- Other manufacturers have all rules in the Data Lims.
- Some Toyota definitions are in CDR "Glossary".
- "**RTFG**" – **R**ead **T**he **F**riggin' **G**lossary"

Toyota EDR Families

First Generation (numbers are calendar year of first use)

- 00/02 MANY record only frontal events
- 00/02 A FEW record side data if side bag equipped, & roll

Second Generation

- 04/06 EDR – MOST have precrash, SOME have side data

Gen 1&2: Precrash data is not tied to a module family.

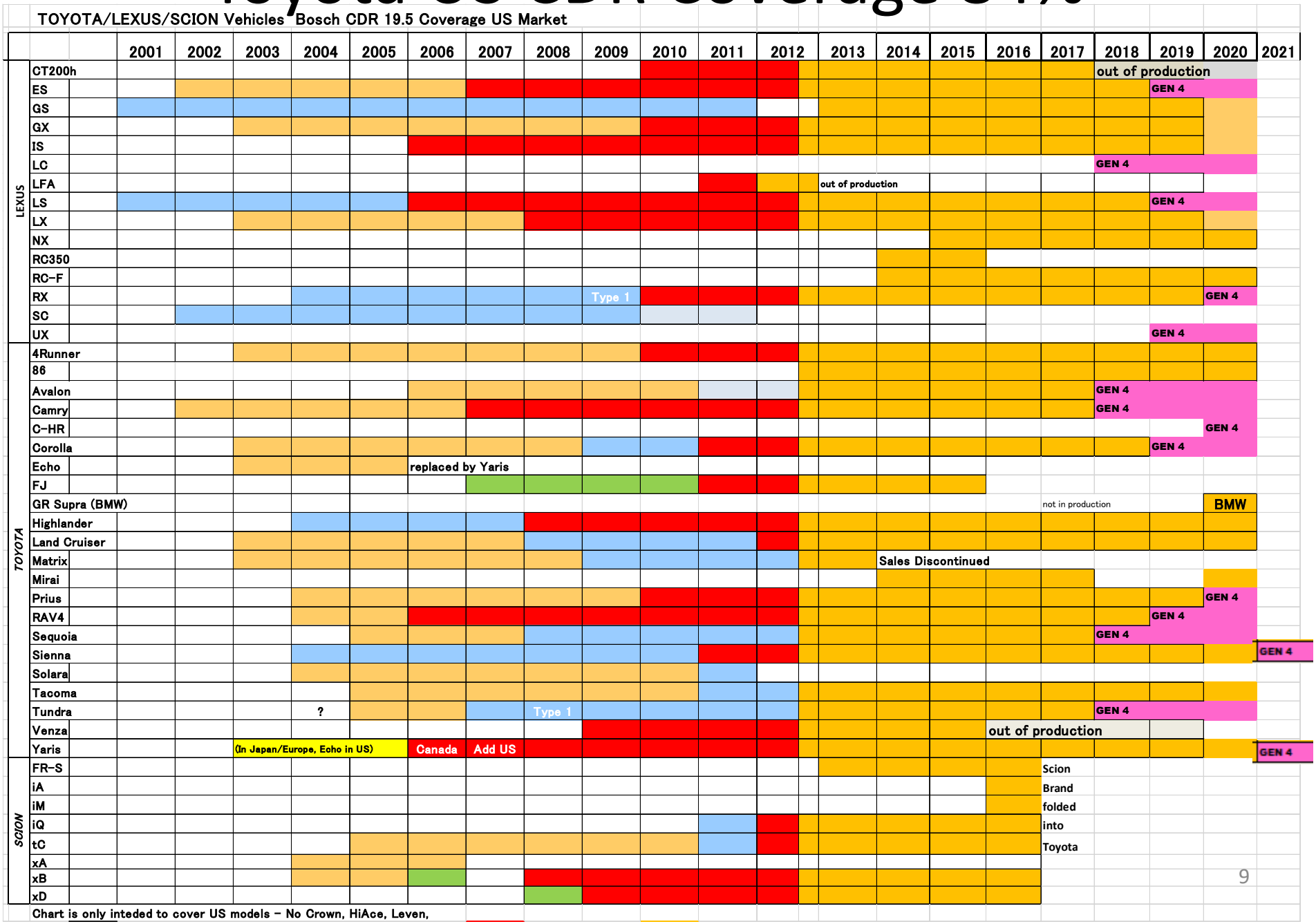
Third Generation

- 12 EDR 563-intent, but still **no** Lat DV in frontal events. If side event is triggered, only 70ms side data from B/C pillar
- 13 EDR, Adds ACM Lat ***accel*** to frontal events
- 15 EDR, adds side impact door pressure sensors

Fourth Generation

- 17 EDR (some 2018+ models) adds Lat ΔV and side triggers to frontal events & optional pop up hood ped protection feature

Toyota US CDR Coverage 84%+



Toyota Unique Help File Organization

Supported Toyota Vehicles

Use this section of the help file to determine if a vehicle is supported by the CDR tool and to look up the correct cable (and in so
note that even if the vehicle is listed in this section, CDR must also support the part number on the ACM. Click here for a list of s

Click here for tips on how to use this *Supported Vehicles* help topic

IMPORTANT: To prevent damage to Toyota ECUs when imaging EDR data directly from the ECU, you must always use the ap

General Toyota Application Notes:

The CDR Tool software will accept most Toyota VINs entered by the user and the *Collect Airbag Control Module Data* icon will n
vehicles listed in the help file below are supported as indicated by the list and any associated coverage notes.

All supported vehicles listed can be imaged through the vehicle's DLC connector using the CDR OBDII DLC Cable (F00K108287)

All supported vehicles require the vehicle's ignition key to be switched on prior to and during EDR record imaging through the vel

Vehicle Specific Application Notes:

Click here to view all vehicle specific Toyota coverage notes

Vehicles Are Listed Alphabetically
Not by Model Year

Toyota Models begin with: 4 8 9 A B C D E F H I K L M N P R S T V W Y Z

4Runner (Toyota) **2003 and newer** **Market: 7**

Year	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
2002 - 2008	ACM	Data - 02EDR	CANplus or CDR 900	CANplus - F00K108287 - *for CANplus interface only, you must use the DLC 4 OBD Adapter when connecting to*	CANplus - (no adapter) Cable#F00K108617 CDR 900 - click here (use listed cable above and no ACM adapter)	Under Center Stack

New Toyota Vehicle Lookup

Camry (Toyota) **2001 and newer** **Market: 7**

7=All Markets the Vehicle is Sold

Year	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
2001 - 2004 (all markets) 2005 (US/Canada)	ACM	Data - 00EDR / 02EDR	CANplus or CDR 900	CANplus - F00K108287 *- for CANplus interface only, you must use the DLC 4 OBD Adapter when connecting to vehicles sold outside the US &* CDR 900 - click here	CANplus - (no adapter) Cable#F00K108615 - (no adapter) Cable#F00K108617 Click here [?] to match the ACM part number to the correct cable CDR 900 - click here (use listed cable above and no ACM adapter)	Under Center Stack

Multiple Cables for same model

Must Look up Cable for DTM by Part No

New Toyota Help File ***U.S.*** *Issues*

- Because the help file applies WORLDWIDE, it lists MANY models not found in the US/CAN.
- Toyotas internal records may be by ***CALENDAR*** YEAR, not ***MODEL*** YEAR. Help File may show EDR is first available in 2003, but that was 2004 Model Year in the US. MOST models are “one year off”.
- Avalon shows EDR in 02 but prior charts showed no EDR until 06 – do not assume EDR present in every vehicle listed in the help file

Unique to Toyota

- Delta V is recorded real time as it is occurring (GM waits until event is over to write anything). Even in older models with no backup power supply for recording, partially written Delta V is possible
- There are multiple “recording complete” entries:
- 1. Complete file
- 2. Crash Pulse subsection (Delta V), per event
- 3. Precrash subsection (Speed), per event

Toyota EDR Evolution

GEN 1 00,02 EDR	GEN 2 04,06 EDR	GEN 3 12,13,15 EDR	GEN 4 17,19 EDR
3 front 1 side 1 roll	2 front 2 side 2 roll		4 front 4 side 2 roll 6 precrash
Record on 2g wakeup		Record on 5mph XDV, side 3.5g wake	Record on 5 X or Y DV
150ms XDV 5PC@1 + TRG	200ms XDV 5PC@1+TRG	250ms XDV 10PC@0.5 + TRG	250ms XY 10PC@0.5+TRG

Toyota EDR Evolution cont'd

GEN 1 00,02 EDR	GEN 2 04,06 EDR	GEN 3 12,13,15 EDR	GEN 4 17,19 EDR
70ms YDV@ACM+sats		12 YDV 70ms sats only 13 250ms ACM	250ms YDV
REFRESH 500 MS SPEED		REFRESH 24 MS SPEED	
7**8.3** MPH TOP, 2kph res	7**5.8** MPH TOP. 2kph res	124.3 MPH/200kph TOP 1 kph res	
Accel Pedal Volts		Accel Pedal % & Throttle %	

Toyota EDR Evolution cont'd

GEN 1
00,02 EDR

GEN 2
04,06 EDR

GEN 3
12,13,15 EDR

GEN 4
17,19 EDR

RPM to next lower 400

RPM to 100

No Key Cycles

Key Cycles

Time Between Events < 5 sec

Time Between Events <16 sec

Brk Oil Pres, Stab. Cont. Long Accel

Strg & Yaw

Unique to Toyota: Record Summary

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash & DTC Data Recording Status	Event & Crash Pulse Data Recording Status
Most Recent Event	7	Side Crash	0	Complete (Page 0)	Complete (Side Page 0)
1st Prior Event	6	Front/Rear Crash	-20	Complete (Page 0)	Complete (Front/Rear Page 1)
2nd Prior Event	5	Side Crash	-228	Complete (Page 0)	Complete (Side Page 1)
3rd Prior Event	4	Front/Rear Crash	-282	Complete (Page 0)	Complete (Front/Rear Page 0)

- Events Trigger (TRG) numbered & time stamped
- 7 events in life of vehicle. Most recent 2 events per crash type are saved regardless of severity.
- For side recording equipped vehicles, in angular crashes, both front and side algorithms will wake up and record data separately. Look for front & side “pairs” with times close to one another.
- Precrash Page 0 is saved at TRG4, not TRG7

00/02 EDR record summary

Front/Rear Event Record Summary at Retrieval Side Events not numbered

Events Recorded	TRG Count	Crash Type	Time (msec)	Event & Crash Pulse Data Recording Status
Most Recent Frontal/Rear Event	4	Front/Rear Crash	0	Complete (Front/Rear Page 1)
1st Prior Frontal/Rear Event	3	Front/Rear Crash	-5120 or greater	Complete (Front/Rear Page 2)
Prior Frontal/Rear Event	1	Front/Rear Crash	N/A	Complete (Front/Rear Page 0)

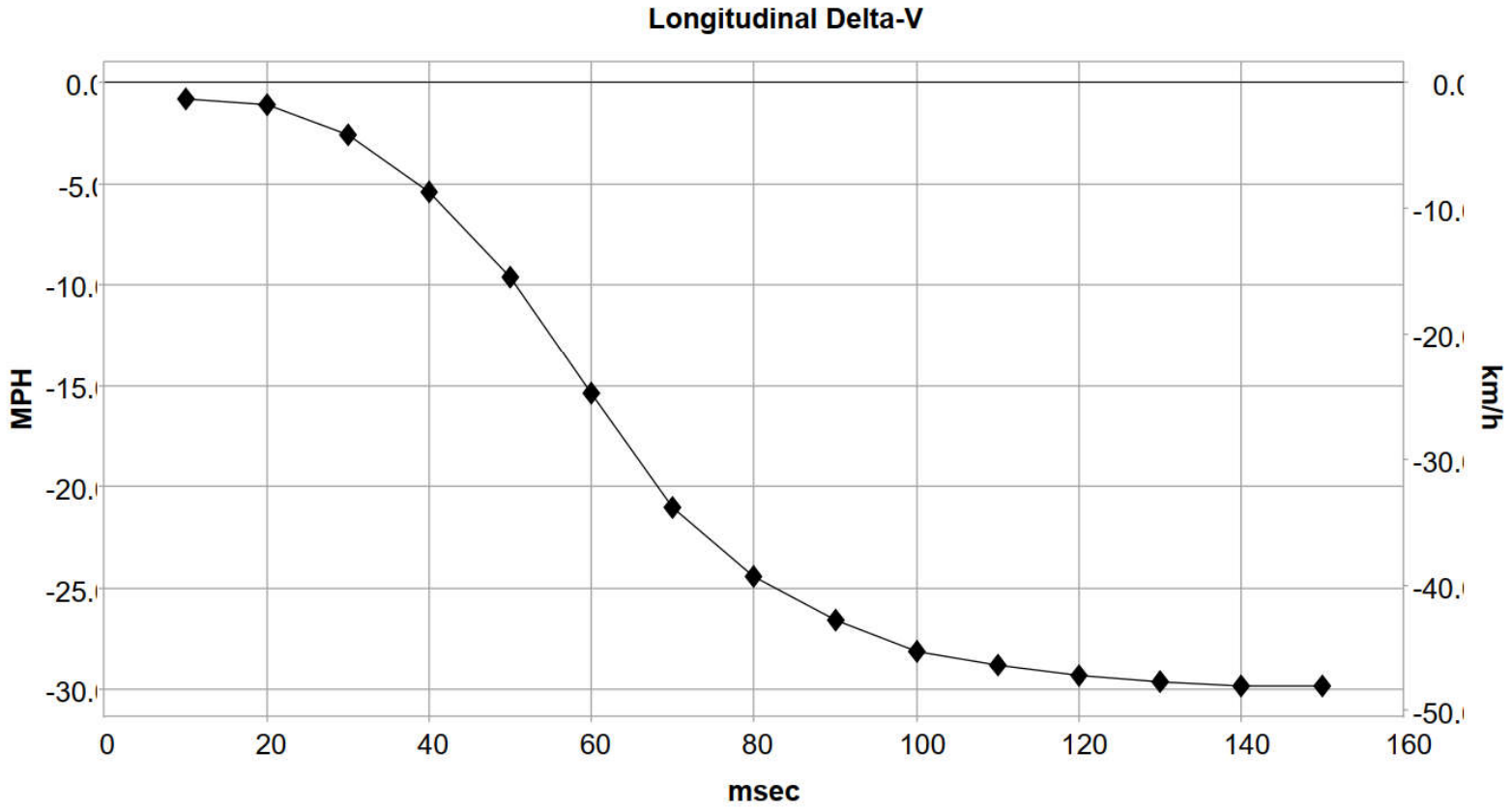
- Saves **third** front/rear event that came closest to deploying airbags (most negative DV).
- 1st prior event #3 = -5120ms ***or greater*** before most recent event, means ***NOT RELATED***
- TRG 2 was less important than TRG 1 and was overwritten when TRG 4 came in.

Gen 1 (and 2) Max Long. DV

- "Max" on top is the biggest number in the data table. Due to shorter duration recording, this is usually also very near the end of the crash.

Longitudinal Crash Pulse (Most Recent Frontal/Rear Event, TRG 2 - table 1 of 2)

Max Longitudinal Delta-V (MPH [km/h])	-29.9 [-48.1]



Finding PDOF in Toyota Gen1&2

- Toyota Gen 1&2 record frontal data when frontal algorithm wakes up, side data **only** when side algorithm wakes up and side data is supported.
- Many early Toyotas do **not** support side recording, you must estimate PDOF from visible crush damage.
- In later Toyotas with side recording, look for “pairs” of front and side recordings. Side recording duration is about 70ms, get PDOF by taking Y value at 70ms and X value from 70ms. BE CAREFUL – Side algorithm may wake up at a different time than frontal, **try to synchronize them.**

Unique to Toyota

- Toyota records ADDITIONAL precrash data point at TRG and the **time** from *last regular interval* recording to TRG (resolution to 0.1 sec, 0.0 to 0.9, truncated down)
- Last data may be WRITTEN at TRG, but could still be "old" as speed data in **Gen 1 & 2** was only transmitted every 0.5 sec. If speed data did not change from prior data point, it may be old by 0.5 sec. If the last reported value is different, it cannot be older than the time from last value to prior value.

Gen 1 & 2 Pre-Crash Data

- Accel Pedal Data in ***Voltage from sensor***
- Zero accel pedal is typically 0.78V, 100% is 3.79 V max seen to date but data limitations indicates the low and high could be different.
- You can calculate your own percentage if you want. %=(V-0.78)/(3.79-0.78) Update .5 sec
- RPM in 400 RPM increments, 400-799=400, 800-1199=800, etc. Updates every 24ms.
- Brake on/off updates immediately, no delay (accel pedal may as on when brake is on, but they are from different times)

Unique Toyota Memory Structure

- Toyota records pre-crash in a different memory section than Delta V. It gives records "page numbers" 0 and 1 and tells you which Precrash page goes with which Delta V page.
- If there are two or more events within 0.5 sec, all may be linked to the same precrash page. This is noted in the event record summary. CDR will print the SAME precrash page 4 times
- Precrash Page 0 is stored at TRG4, not TRG7

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash & DTC Data Recording Status	Event & Crash Pulse Data Recording Status
Most Recent Event	7	Side Crash	0	Complete (Page 0)	Complete (Side Page 0)
1st Prior Event	6	Front/Rear Crash	-20	Complete (Page 0)	Complete (Front/Rear Page 1)
2nd Prior Event	5	Side Crash	-228	Complete (Page 0)	Complete (Side Page 1)
3rd Prior Event	4	Front/Rear Crash	-282	Complete (Page 0)	Complete (Front/Rear Page 0)

Un-linked precrash data

- The remaining precrash space is then known as “UNLINKED PRE CRASH” and was likely linked to a previous event. It is printed after the last linked precrash page
- In multiple event scenarios, Unlinked precrash data overlaps linked precrash and shows driver behavior farther back in time. Always check for overlap!

System Status at Time of Retrieval

ECU Part Number	89170-52D80
ECU Generation	06EDR
Recording Status, All Pages	Complete
Freeze Signal	ON
Freeze Signal Factor	Front Airbag Deployment Front Pretensioner Deployment
Contains Unlinked Pre-Crash Data	Yes

Overlay Unlinked Data with Linked

Pre-Crash Data, -5 to 0 seconds (1st Prior Event, TRG 8)

Time (sec)	-4.1	-3.1	-2.1	-1.1	-0.1	0 (TRG)
Vehicle Speed (MPH [km/h])	74.6 [120]	70.8 [114]	70.8 [114]	69.6 [112]	36 [58]	36 [58]
Brake Switch	ON	OFF	OFF	OFF	OFF	OFF
Accelerator Rate (V)	0.78	0.78	1.05	1.45	0.78	0.78
Engine RPM (RPM)	2,400	2,400	3,200	3,200	1,200	800

Pre-Crash Data, -5 to 0 seconds (Unlinked, Page 1)

Time (sec)	-4.7	-3.7	-2.7	-1.7	-0.7	0 (TRG)
Vehicle Speed (MPH [km/h])	75.8 [122]	75.8 [122]	74.6 [120]	70.8 [114]	70.8 [114]	69.6 [112]
Brake Switch	ON	ON	ON	OFF	OFF	OFF
Accelerator Rate (V)	0.78	0.78	0.78	0.78	1.05	1.56
Engine RPM (RPM)	3,600	2,800	2,400	2,400	3,200	3,200

These two precrash data sets are from the same crash sequence

Delta V Data in Gen 2 models

- Longer Duration Longitudinal DV (200ms vs 150ms)

  20 points @10ms intervals = 200ms duration starts at algorithm wake up - enough to get most crashes.
- Lateral DV (if equipped) for B pillar, C pillar and “Floor” (ACM) locations 25 points @4ms intervals – 100 ms duration starting 20ms before wakeup. Only use ACM (floor) location for recon.
- Roll Angle & Lateral Accel (if equipped).

  20 points @ 125 ms intervals beginning 625 ms before wake up

  Total duration 2.4 seconds
- Graph vs time, data table, and “Max” single value
- Same guidelines for use as Toyota Gen 1

Other Useful Recon Data

- Front Pass Seat Belt Buckle status
- Passenger Weight Classification (Empty/Unoccupied, Child, AF05 (small adult), **AM 50 (midsize adult *or larger*).** (*Many others only have small adult or larger class*).
- Driver Seat Position (FW forward vs RW)
- Gearshift lever position (P-R-N-Drive)
- What safety devices deployed when, deploy time marked on Delta V graph.

Toyota Gen 2 Front Crash Report

BOSCH

CDR CRASH DATA RETRIEVAL SYSTEM

IMPORTANT NOTICE: Robert Bosch LLC and the manufacturers whose vehicles are accessible using the CDR System urge end users to use the latest production release of the Crash Data Retrieval system software when viewing, printing or exporting any retrieved data from within the CDR program. Using the latest version of the CDR software is the best way to ensure that retrieved data has been translated using the most current information provided by the manufacturers of the vehicles supported by this product.

CDR File Information

User Entered VIN	YOUR VIN HERE
User	NONE
Case Number	NONE
EDR Data Imaging Date	
Crash Date	
Filename	52690 POLICE 38.ROT.CDRX
Saved on	
Collected with CDR version	Crash Data Retrieval Tool 4.0
Reported with CDR version	Crash Data Retrieval Tool 4.0
EDR Device Type	airbag control module
Event(s) recovered	Front/Rear (2)

Events: Front/Rear vs Side vs Rollover - & how many

Comments

Data imported from: 52690 POLICE 38.ROT

Data Limitations

CDR Record Information:

- Due to limitations of the data recorded by the airbag ECU, such as the resolution, data range, sampling interval, time period of the recording, and the items re
- Airbag ECU data should b
circumstances.
- If the airbags did not deplo called a Non-Deployment Event. D ecording threshold. If the airbag(s) deploy or the pretensioners are operated, it is called a Deployment Event. Deployment Event data cannot be overwritten or deleted by the airbag ECU following that event

P. 1 Same Look/Feel as others

System Status at Retrieval

System Status at Time of Retrieval

ECU Part Number	89170-52690
ECU Generation	04EDR
Recording Status, All Pages	Complete
Freeze Signal	ON
Freeze Signal Factor	Front Airbag Deployment Front Pretenstioner Deployment
Diagnostic Codes Exist	No
Total number of Events	2
Time from Previous Pre Crash TRG (msec)	16381 or greater
Latest Pre-Crash Page	0
Contains Unlinked Pre-Crash Data	No

Means record is locked, no other events can be recorded

Why record is locked

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash & DTC Data Recording Status	Event & Crash Pulse Data Recording Status
Most Recent	2	Front/Rear Crash	0	Complete (Page 0)	Complete (Front/Rear Page 1)
1st Prior	1	Front/Rear Crash	-200	Complete (Page 0)	Complete (Front/Rear Page 0)

Same Precrash Page

Back to back events

System Status at Event

System Status at Event (Most Recent, TRG 2)

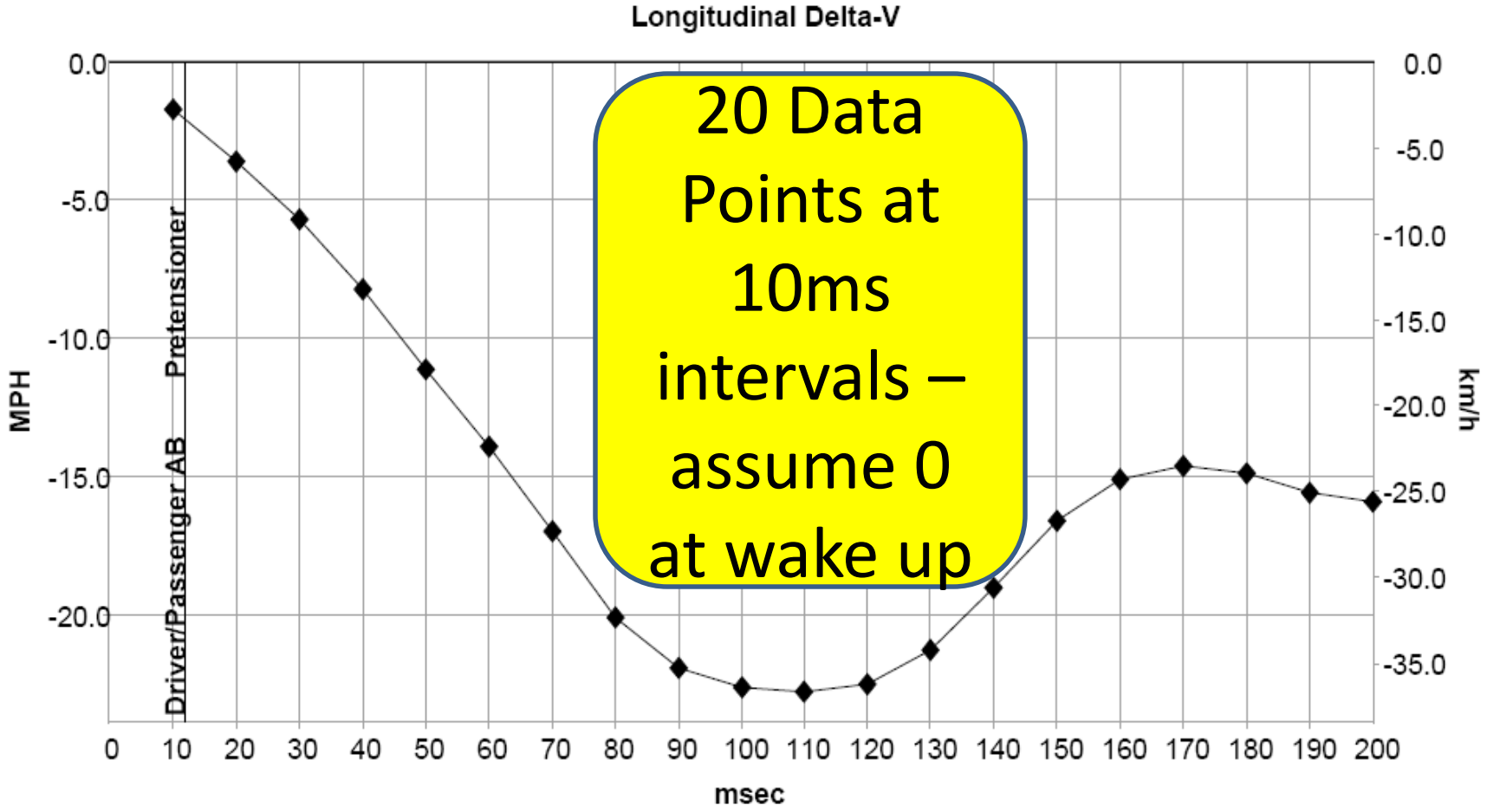
Recording Status, Front/Rear Crash Info.	Complete
Event Type	Front/Rear Crash
TRG Count (times)	2
Previous Crash Type	Frontal/Rear
Time from Pre-Crash TRG (msec)	200
Linked Pre-Crash Page	0
Time to Deployment Command, Front Airbag, Driver (msec)	12
Time to Deployment Command, Front Airbag, Passenger (msec)	12
Event Severity Status, Driver	Level 3
Event Severity Status, Passenger	N/A
Time to Deployment Command, Pretensioner (msec)	12

Timing between airbag stages 3=short 1=long

Delta V Data Graph

Longitudinal Crash Pulse (Most Recent, TRG 2 - table 1 of 2)

Recording Status, Time Series Data	Complete
Max Longitudinal Delta-V (MPH[km/h])	-22.8 [-36.7]



Delta V Data Table

Longitudinal Crash Pulse (Most Recent, TRG 2

Time (msec)	Longitudinal Delta-V (MPH [km/h])
10	-1.7 [-2.7]
20	-3.6 [-5.8]
30	-5.7 [-9.2]
40	-8.2 [-13.2]
50	-11.1 [-17.9]
60	-13.9 [-22.4]
70	-17.0 [-27.4]
80	-20.1 [-32.3]
90	-21.9 [-35.2]
100	-22.6 [-36.4]
110	-22.8 [-36.7]
120	-22.5 [-36.2]
130	-21.3 [-34.3]
140	-19.0 [-30.6]
150	-16.6 [-26.7]
160	-15.1 [-24.3]
170	-14.6 [-23.5]
180	-14.9 [-24.0]
190	-15.6 [-25.1]
200	-15.9 [-25.6]

20 Data Points at 10ms intervals – start at wake-up (assume 0 at 0ms)

Pre Crash Data - Overview

Same format Gen 1 and Gen 2

DTCs Present at Time of Event (Most Recent, TRG 2)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Some have default max value 2047

Some have default max value 15

Pre-Crash Data, 1 Sample (Most Recent, TRG 2)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	200
Buckle Switch, Driver	Unbuckled
Buckle Switch, Passenger	Unbuckled
Occupancy Status, Passenger	Not Occupied
Seat Position, Driver	Rearward
Shift Position	Drive

Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.2	-3.2	-2.2	-1.2	-0.2	0 (TRG)
Vehicle Speed (MPH [km/h])	12.4 [20]	8.7 [14]	5 [8]	2.5 [4]	1.2 [2]	1.2 [2]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	800	800	800	400	800	800

Pre Crash Data Timing

Other Points a regular 1 sec intervals

6th point Sampled at TRG

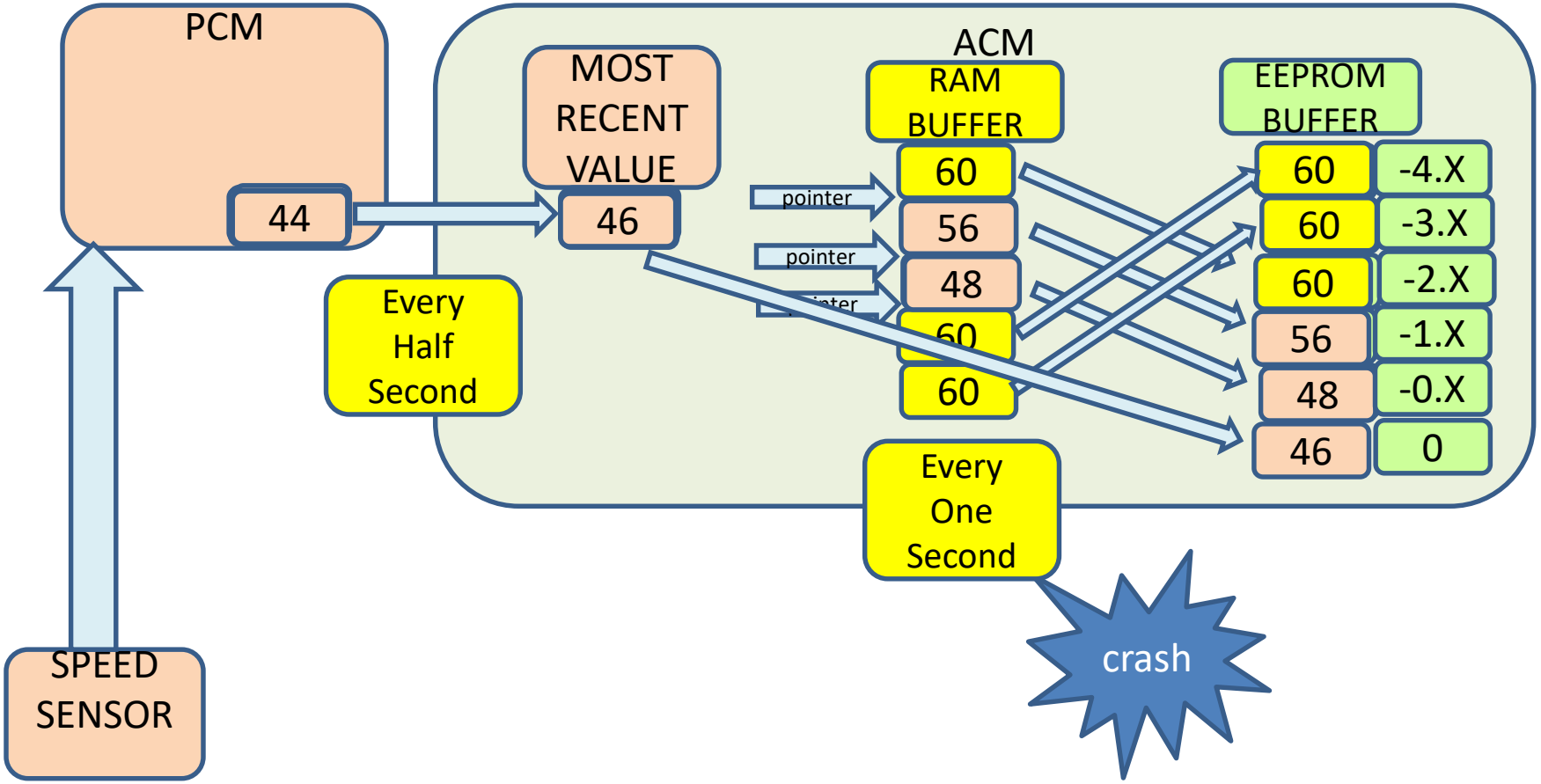
Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.2	-3.2	-2.2	-1.2	-0.2	0 (TRG)
Vehicle Speed (MPH [km/h])	12.4 [20]	8.7 [14]	5 [8]	2.5 [4]	1.2 [2]	1.2 [2]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	800	800	800	400	800	800

Caution: Sampling at TRG could have un-refreshed data in buffer

New Pre Crash data will only be recorded if 500ms have passed since the last event.

Toyota speed data transfer[34]



Gen 1 & 2 Pre Crash Data Timing

Other Points a 1 sec intervals

6th point Sampled at TRG

Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.2	-3.2	-2.2	-1.2	-0.2	0 (TRG)
Vehicle Speed (MPH [km/h])	12.4 [20]	8.7 [14]	5 [8]	2.5 [4]	1.2 [2]	1.2 [2]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	800	800	800	400	800	800

Caution: Sampling at TRG could have un-refreshed data in buffer

New Pre Crash data will only be recorded if 500ms have passed since the last event.

Gen 1 & 2 Pre Crash Speed Data

Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.2	-3.2	-2.2	-1.2	-0.2	0 (TRG)
Vehicle Speed (MPH [km/h])	12.4 [20]	8.7 [14]	5 [8]	2.5 [4]	1.2 [2]	1.2 [2]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	800	800	800	400	800	800

- Since the speed data buffer is only refreshed every 0.5 sec, the 0 and -0.2 data points may be from the same buffer value. If the time between TRG and the prior point is less than 0.5 sec and the values are identical, especially during precrash braking, do not assume the vehicle has stopped slowing down.
- Resolution 2km/hr (1.2 mph)
- Max Value 75.8 mph (122kph) on Gen 2 (04/06), (78.3mph/126kph in Gen 1 00/02 models)

Gen 1 & 2 Pre Crash Accel Pedal Data

Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.2	-3.2	-2.2	-1.2	-0.2	0 (TRG)
Vehicle Speed (MPH [km/h])	12.4 [20]	8.7 [14]	5 [8]	2.5 [4]	1.2 [2]	1.2 [2]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	800	800	800	400	800	800

- Only refreshed every 512 ms – held in buffer
- Volts not pedal percent
- Convert to % = (V-0.78)/(3.79-0.78)
- Data Limitations warns min & max may vary

Gen 1 & 2 Pre Crash RPM Data

Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.2	-3.2	-2.2	-1.2	-0.2	0 (TRG)
Vehicle Speed (MPH [km/h])	12.4 [20]	8.7 [14]	5 [8]	2.5 [4]	1.2 [2]	1.2 [2]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	800	800	800	400	800	800

- Resolution to next lower 400 rpm. Max value 5200 rpm
- If speed is reported at max 75 mph, can be used to calculate a higher min speed based on gear ratios.
- Speed in mph = RPM/trans ratio/final drive ratio/tire revs /mile*60

- Next Event

System Status at Event (1st Prior, TRG 1)

Recording Status, Front/Rear Crash Info.	Complete
Event Type	Front/Rear Crash
TRG Count (times)	1
Previous Crash Type	No Event
Time from Pre-Crash TRG (msec)	0
Linked Pre-Crash Page	0
Time to Deployment Command, Front Airbag, Driver (msec)	Not Commanded
Time to Deployment Command, Front Airbag, Passenger (msec)	Not Commanded
Event Severity Status, Driver	N/A
Event Severity Status, Passenger	N/A
Time to Deployment Command, Pretensioner (msec)	Not Commanded

RTFDL Caution

- In the 00 EDR, seat belt status in ND events is the status at time of READOUT, not the time of EVENT.
- This is duly noted in the data limitations, but has tripped some people up. A module was read in car that said unbelted, then re-read on the benchtop that gave the default value of belted.

- For Non-Deployment Event, "Buckle Switch", "Ignition Cycle Since DTC was Set" and "Airbag Warning Lamp ON Time Since DTC was Set" is the ACM parameter data at the time of retrieval, not the parameter data when the recording trigger was established. For Deployment Event, it is the information when the airbag deployment.

Bug in Gen 1&2 Long. Delta V

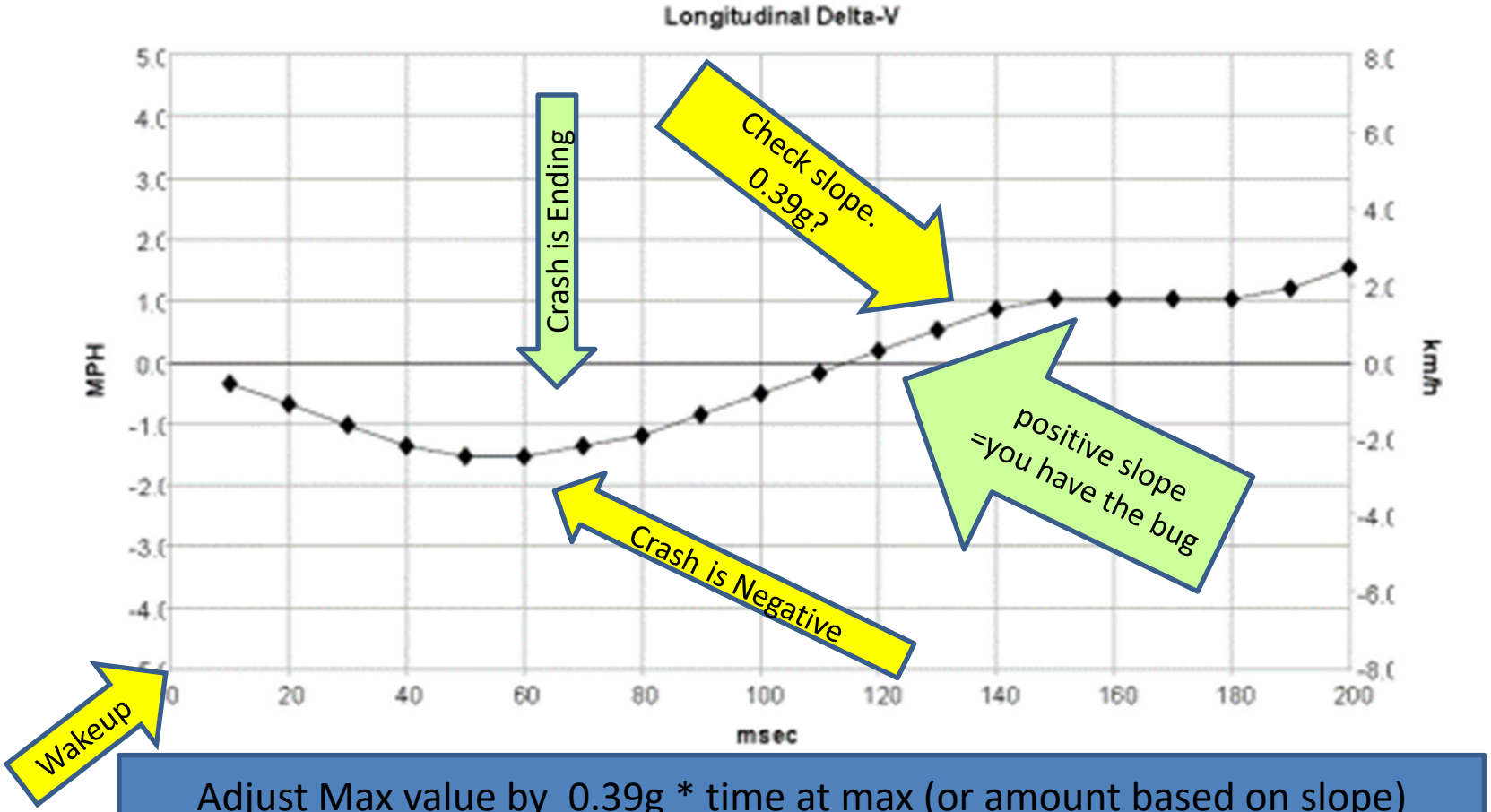
- Empirical research indicates in ***all Gen 1's*** tested to date and ***SOME Gen 2*** models
- ***Under*** reports *frontals* slightly
- ***Over*** reports *rears* slightly
- Sources in Toyota indicate the zero-offset applied when converting analog to digital accelerometer output may be off by 1 count, creating an error of 0.39G.
- 0.39g*150msec*32.2/1.466 = 1.3 mph
- Empirical data suggests 2 counts on some
- SAE papers from MEA discussing this include

  2017-01-1438, 2016-01-1494, 2013-01-1268

How to Identify you have the bug

Longitudinal Crash Pulse (Most Recent Event, TRG 3 - table 1 of 2)

Recording Status, Time Series Data	Complete
Max Longitudinal Delta-V (MPH [km/h])	1.5 [2.5]



Adjust Max value by 0.39g * time at max (or amount based on slope)

No positive slope = no adjustment needed

Toyota Gen 2 Side Crash Report

BOSCH

CDR CRASH DATA RETRIEVAL SYSTEM

IMPORTANT NOTICE: Robert Bosch LLC and the manufacturers whose vehicles are accessible using the CDR System urge end users to use the latest production release of the Crash Data Retrieval system software when viewing, printing or exporting any retrieved data from within the CDR program. Using the latest version of the CDR software is the best way to ensure that retrieved data has been translated using the most current information provided by the manufacturers of the vehicles supported by this product.

CDR File Information

User Entered VIN	4T1BF3EK4BU664098
User	Richard R. Ruth, P.E.
Case Number	Gazdick salvage yard find
EDR Data Imaging Date	06/01/2011
Crash Date	
Filename	4T1BF3EK4BU664098_ACM_GAZDICK_BODY_SHOP.CDRX
Saved on	Wednesday, June 1 2011 at 21:52:48
Collected with CDR version	Crash Data Retrieval Tool 4.0
Reported with CDR version	Crash Data Retrieval Tool 4.0
EDR Device Type	airbag control module
Event(s) recovered	Front/Rear (1), Side (1)

Comments

imaged direct to module with cable 614 and 387 adaptor

Data Limitations

CDR Record Information:

- Due to limitations of the data recorded by the airbag ECU, such as the resolution, data range, sampling interval, time period of the recording, and the items recorded, the information provided by this data may not be sufficient to capture the entire crash.
- Airbag ECU data should be used in conjunction with other physical evidence obtained from the vehicle and the surrounding circumstances.
- If the airbags did not deploy or the pretensioners did not operate during an event that meets a specified recording threshold, it is called a Non-Deployment Event. Data from a Non-Deployment Event can be overwritten by a succeeding event that meets the specified recording threshold. If the airbag(s) deploy or the pretensioners are operated, it is called a Deployment Event. Deployment Event data cannot be overwritten or deleted by the airbag ECU following that event

Events: Front/Rear vs Side vs Rollover - & how many

Side Event Polarity

Which Side were you hit on?

System Status at Event (Most Recent, TRG 2)

Recording Status, Side Crash Info.	Complete
Event Type	Side Crash
TRG Count (times)	2
Recorded Side	Passenger's Side
Previous Crash type	Frontal/Rear
Time from Pre-Crash TRG (msec)	24
Linked Pre-Crash Page	0
Time to Deployment Command, Row 1 (msec)	Not Commanded
Time to Deployment Command, Row 2 (msec)	Not Commanded

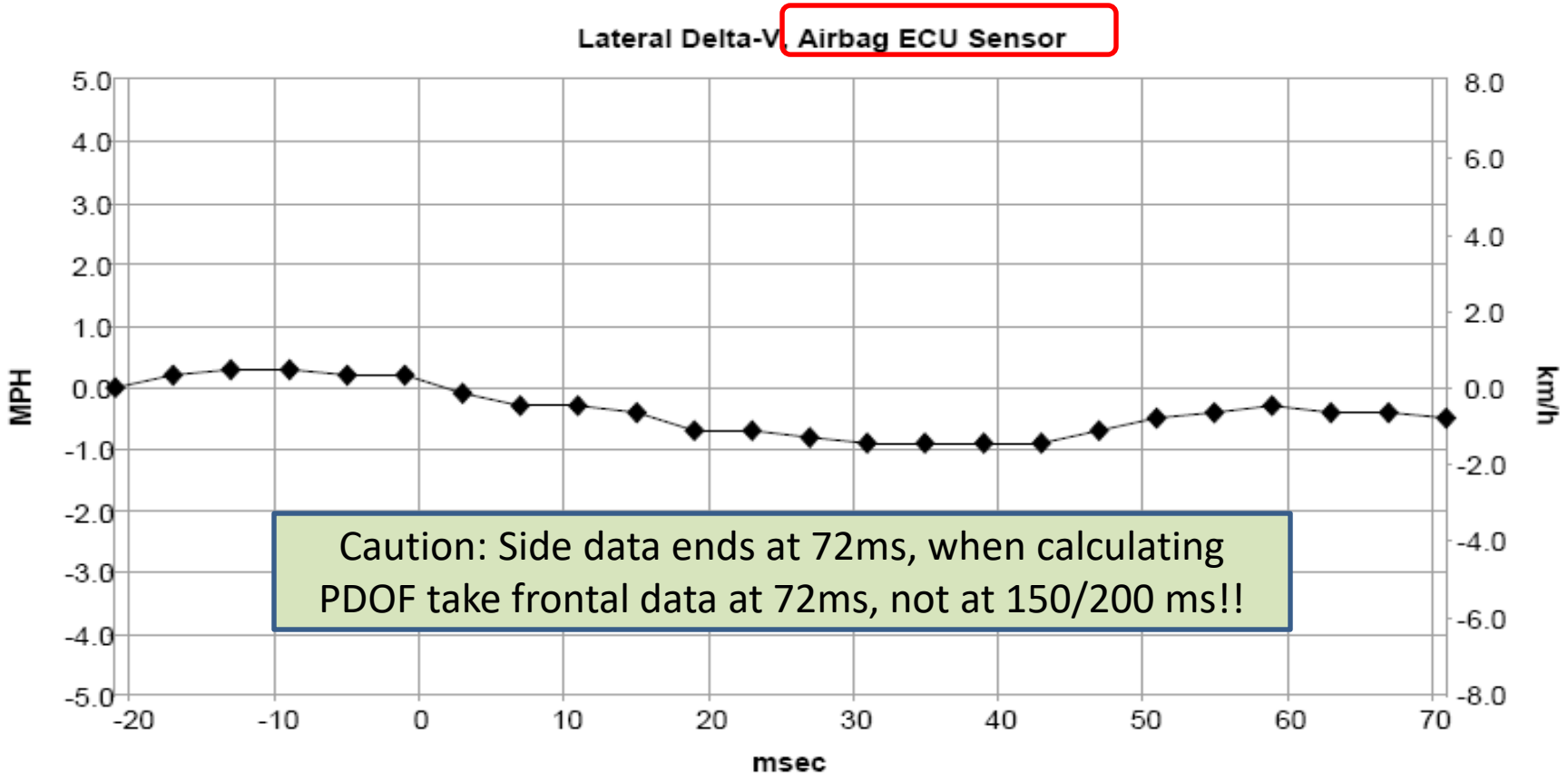
- Side contacted is NOT determined by sign.
- Recorded Side is displayed.

Gen 1&2: Lateral Delta V Graph

Floor Sensor means ACM (same as Airbag ECU) – closest to CG

Lateral Crash Pulse (Most Recent, TRG 2 - table 1 of 2)

Recording Status, Time Series Data	Complete
Time from TRG to Next Sample (msec)	3
Max Lateral Delta-V, Floor (MPH[km/h])	-0.9 [-1.4]
Max Lateral Delta-V, B-Pillar Sensor (MPH[km/h])	1.5 [2.4]
Max Lateral Delta-V, C-Pillar Sensor (MPH[km/h])	2.7 [4.3]



Gen 1&2: Use ECU Side Data Only

Lateral Crash Pulse (Most Recent, TRG 2 - table 2 of 2)

Time (msec)	Lateral Delta-V, Airbag ECU Sensor (MPH [km/h])	Lateral Delta-V, B-Pillar Sensor	Lateral Delta-V, C-Pillar Sensor
-21	0.0 [0.0]		
-17	0.2 [0.3]		
-13	0.3 [0.5]		
-9	0.3 [0.5]		
-5	0.2 [0.3]		
-1	0.2 [0.3]		
3	-0.1 [-0.2]		
7	-0.3 [-0.5]		
11	-0.3 [-0.5]		
15	-0.4 [-0.6]		
19	-0.7 [-1.1]		
23	-0.7 [-1.1]		
27	-0.8 [-1.3]		
31	-0.9 [-1.4]		
35	-0.9 [-1.4]		
39	-0.9 [-1.4]		
43	-0.9 [-1.4]		
47	-0.7 [-1.1]		
51	-0.5 [-0.8]	1.5 [2.4]	.6 [4.2]
55	-0.4 [-0.6]	1.5 [2.4]	2.7 [4.3]
59	-0.3 [-0.5]	1.5 [2.4]	2.7 [4.3]
63	-0.4 [-0.6]	1.4 [2.3]	2.7 [4.3]
67	-0.4 [-0.6]	1.4 [2.3]	2.7 [4.3]
71	-0.5 [-0.8]	1.5 [2.4]	2.7 [4.3]

Pre Crash Data Page Gen 1& 2 (looks **same** for side and frontal)
Check page numbers, if page # is same they are identical

DTCs Present at Time of Event (Most Recent, TRG 2)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (Most Recent, TRG 2)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	700
Buckle Switch, Driver	Buckled
Buckle Switch, Passenger	Unbuckled
Occupancy Status, Passenger	Not Occupied
Seat Position, Driver	Rearward
Shift Position	Drive

Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.7	-3.7	-2.7	-1.7	-0.7	0 (TRG)
Vehicle Speed (MPH [km/h])	46 [74]	47.2 [76]	47.2 [76]	47.2 [76]	42.3 [68]	24.9 [40]
Brake Switch	OFF	OFF	OFF	OFF	ON	ON
Accelerator Rate (V)	1.25	1.09	0.98	0.98	0.78	0.78
Engine RPM (RPM)	1,600	1,200	1,200	1,200	800	800

Speed at Impact Worksheet GEN 1 & 2

TWO changes from 42 to 24?

PRE-CRASH DATA						
Time	-4.7	-3.7	-2.7	-1.7	-0.7	0
Speed	46 (74)	47.2 (76)	47.2 (76)	47.2 (76)	42.3 (68)	24.9 (40)
Brake	Off	Off	Off	Off	On	On
Time Range	-5.2 to -4.7	-4.2 to -3.7	-3.2 to -2.7	-2.2 to -1.7	-1.2 to -0.7	0.2 to 0.0

SPEED AT IMPACT WORKSHEET		English		Metric	
		MIN	MAX	MIN	MAX
Last Good Data Point mph (km/h)		24.9	24.9	40	40
Braking	-0.2 sec* 15 mph/sec	-3.0	0	-4.8	0
Slip +5%		1.2	1.2	2	2
Speedo +/-4%		-1	1	-1.6	1.6
Range		22.1	27.1	35.6	43.6

Age of TRG data sample Gen 1 & 2

Time Last Point to TRG	Changed twice	Changed once	Didn't change
0	n/a	0	0.5
0.1	n/a	0.1	0.5
0.2	n/a	0.2	0.5
0.3	n/a	0.3	0.5
0.4	n/a	0.4	0.5
0.5	0	0.5	n/a
0.6	0.1	0.5	n/a
0.7	0.2	0.5	n/a
0.8	0.3	0.5	n/a
0.9	0.4	0.5	n/a

- Data at TRG can be 0.5 sec old, but may be LESS than 0.5 sec if data has changed ONCE from prior sample within 0.5 sec (9 mph or less)
- Data at TRG can ALSO be LESS than 0.5 sec if data has changed TWICE from prior sample (10 mph or more)
- Version 2.01 of the speed at impact worksheet excel program does this for you

Gen 3 -Part 563 Intent Toyota”12EDR”

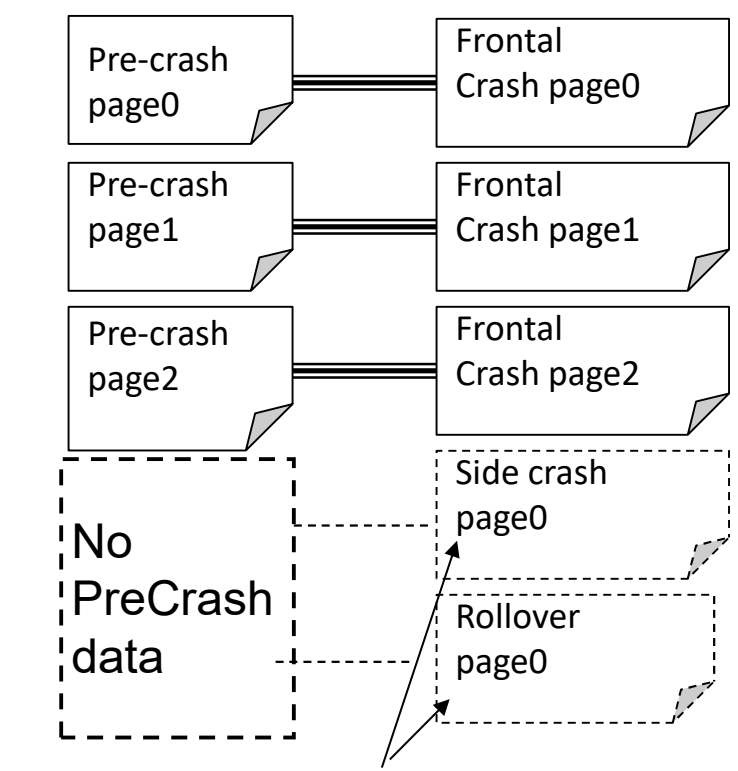
- Phase in mid 2012 MY
- Adds key cycles, deploy times, longer ΔV (250ms), more pre-crash 5 sec @ 0.5 sec intervals, speed to 124 mph (200 KPH), Accel %.
- Lat ΔV only recorded in side events (Unique to Toyota) and ONLY FROM SIDE SATELLITES, not ACM
- Recording threshold 5mph ΔV min. for frontal event
- Side recording threshold stays at 3.5g wakeup
- Adds Throttle %, Steering, Yaw Rate, Brake Pressure, VSC Long Accel, Cruise, Gear, RPM @ 100 rpm resolution.

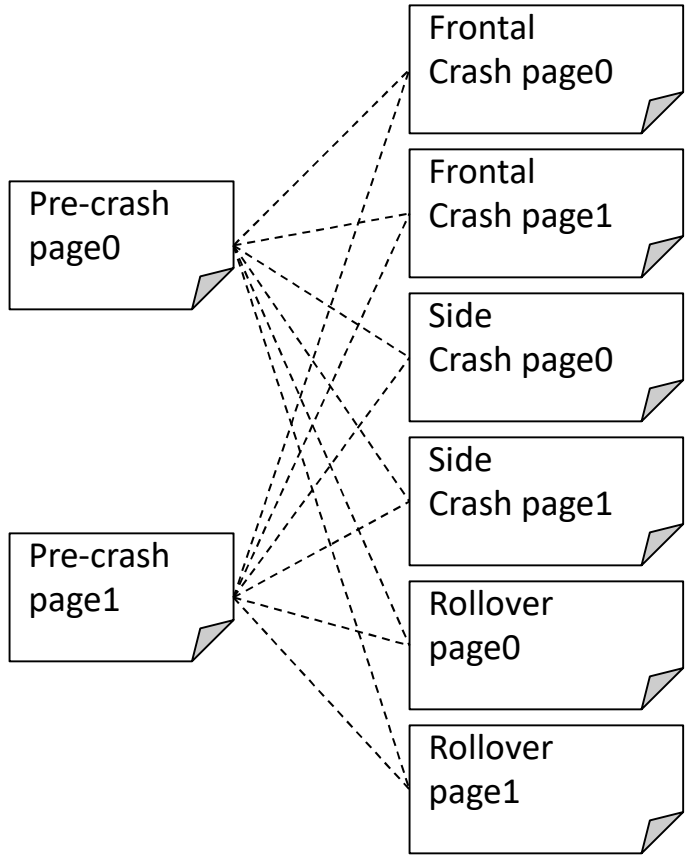
Gen 3 “12 EDR”

- Speed data updated more frequently now - every 20 to 24 ms – (64 ms on accel pedal)
- Toyota continues to sample data point at TRG
- Brake **pressure** data element added in MPa
- Accel pedal data comeS in % vs. volts
- Analyze the same as any other 563 intent module, except recognize that last data point is for all practical purposes **at TRG** now.

Toyota Memory "Pages"

00/02EDR (GEN 1)	04/06EDR (GEN 2) and **12EDR (GEN 3)**
▪Frontal Crash: **3** pages ▪Side and Rollover: 1page (no Pre-Crash)	▪Pre-Crash : **2** pages ▪Frontal Side Rollover :2 pages each





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Pre-Crash Parameter Comparison

Parameter	04/06 EDR		12 EDR	
Vehicle Speed (Part 563)	2 km/h /LSB	0~122 [km/h]	1 km/h /LSB	0 ~ 200 [km/h]
Accelerator Pedal (Part 563)	.039 V/LSB	0~5 [V]	0.5 %/LSB	0 ~ 100 [%]
Engine RPM (Part 563)	400 RPM/LSB	0~5200 [RPM]	100 RPM/LSB	0 ~ 12800 [RPM]
Service Brake (Part 563)	ON/OFF		ON/OFF	
Shift Position	R / N/ P / D		R / N / P / D / 5 / 4/ 3 / 2/ LO/ B/ Seq	
Percentage of Engine Throttle			0.5%/LSB	0 ~ 100 [%]
Motor RPM (Part 563)			100 RPM/LSB	-7500 ~ 17500 [RPM]
Brake Oil Pressure			0.048 MPa	0 ~ 12.144 [MPa]
Sequential Shift			1 / 2 / 3 /4 / 5 / 6 / 7 / 8	
Steering Input (Part 563)			3 deg/LSB	-375 ~ 375 [deg]
VSC Yaw Rate			0.488 deg/s/LSB	-61 ~ 61 [deg/s]
VSC Longitudinal Acceleration			0.07178 m/s^2/LSB	-8.9725~8.9725 [m/s^2]
Cruise Control Status			ON/OFF	
Drive Mode - Power Drive Mode – Eco Drive Mode - Sport Drive Mode - Snow Drive Mode - EV			ON/OFF	

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Gen 3 Longitudinal Stability Control Accel Data

- Units are in m/sec^2, divide by 9.8 to get G's (dividing by 10 gets you close)
- Data is like a Vericom, similar to Ford stability control system data, just not as frequent.
- The max value is 8.973 m/sec^2.
- The last value at TRG is often 8.973, it may be sampled as the crash is beginning. Do not assume max braking from last data point alone.

Getting PDOF from 12 EDR

- 12 EDR records side Delta V ONLY if side algorithm wakes up, and ONLY from side satellites – NO data from ACM!!!!!
- In Gen 1&2, we told you to IGNORE B and C pillar data and use ACM data only.
- With NO DATA from ACM recorded, your only choice is to use B pillar data (closest sensor to CG)
- Take side Delta V at end of trace (typically 70ms) and take arc tangent of Y over X at same time (70ms).
- Try to synchronize X and Y pulses based on relative wake-up

New “13 EDR” & “15 EDR”

- 13 EDR Adds lateral Accel data from ACM (“floor sensor”) in ***front/rear*** **events, synchronized to Long DV time 0** (Fixes biggest drawback to 12 EDR),
- Lateral acceleration data table is in meters/sec^2, calculate Delta V from it yourself. For each point, take **(accel/9.81)*(32.2/1.466)*time interval**, and add them to get the cumulative value in **mph**.
- Metric, take accel*3.6*time interval, add them to get the cumulative value in kph.
- See “Toyota 13 EDR” **TEMPLATE** in student materials to do this math for you calculates both Delta V and PDOF.
- Side events are still recorded separately in side wakeups, & record 70ms of side accel, but are NO LONGER NEEDED for analysis if the frontal algorithm triggered and recorded with side data. .
- 15 EDR adds side impact door pressure sensor. Pressure sensor data is NOT used in reconstruction analysis, it is just for quicker side airbag activation.

13 EDR Long DV & Lat Accel

Longitudinal/Lateral Crash Pulse (3rd Prior Event, TRG 1 - table 2 of 2)

Time (msec)	Longitudinal Delta-V (MPH [km/h])	Lateral Acceleration for Frontal/Rear Crash, Floor Sensor (m/sec^2)	
0	0.0 [0.0]	0.0	
10	-1.1 [-1.7]	4.6	ON
20	-2.0 [-3.3]	7.7	ON
30	-6.6 [-10.6]	41.4	ON
40	-7.8 [-12.5]	96.5	ON
50	-8.6 [-13.8]	19.9	ON
60	-9.1 [-14.7]	-79.7	ON
70	-9.8 [-15.8]	18.4	ON
80	-11.3 [-18.1]	-16.9	ON
90	-12.8 [-20.6]	64.4	ON
100	-13.8 [-22.2]	23.0	ON
110	-15.5 [-25.0]	30.6	ON
120	-17.6 [-28.3]	-49.0	ON
130	-19.8 [-31.8]	13.8	ON
140	-22.7 [-36.5]	24.5	ON
150	-22.3 [-35.9]	-76.6	ON
160	-23.3 [-37.5]	52.1	ON
170	-24.0 [-38.6]	6.1	ON
180	-24.7 [-39.8]	-13.8	ON
190	-25.6 [-41.2]	-12.3	ON
200	-26.1 [-42.1]	9.2	ON
210	-27.0 [-43.5]	29.1	ON
220	-27.5 [-44.2]	26.0	ON
230	-27.7 [-44.6]	19.9	ON
240	-27.9 [-44.9]	7.7	ON
250	-28.1 [-45.2]	4.6	ON

Save as CSV
Integrate to get DV

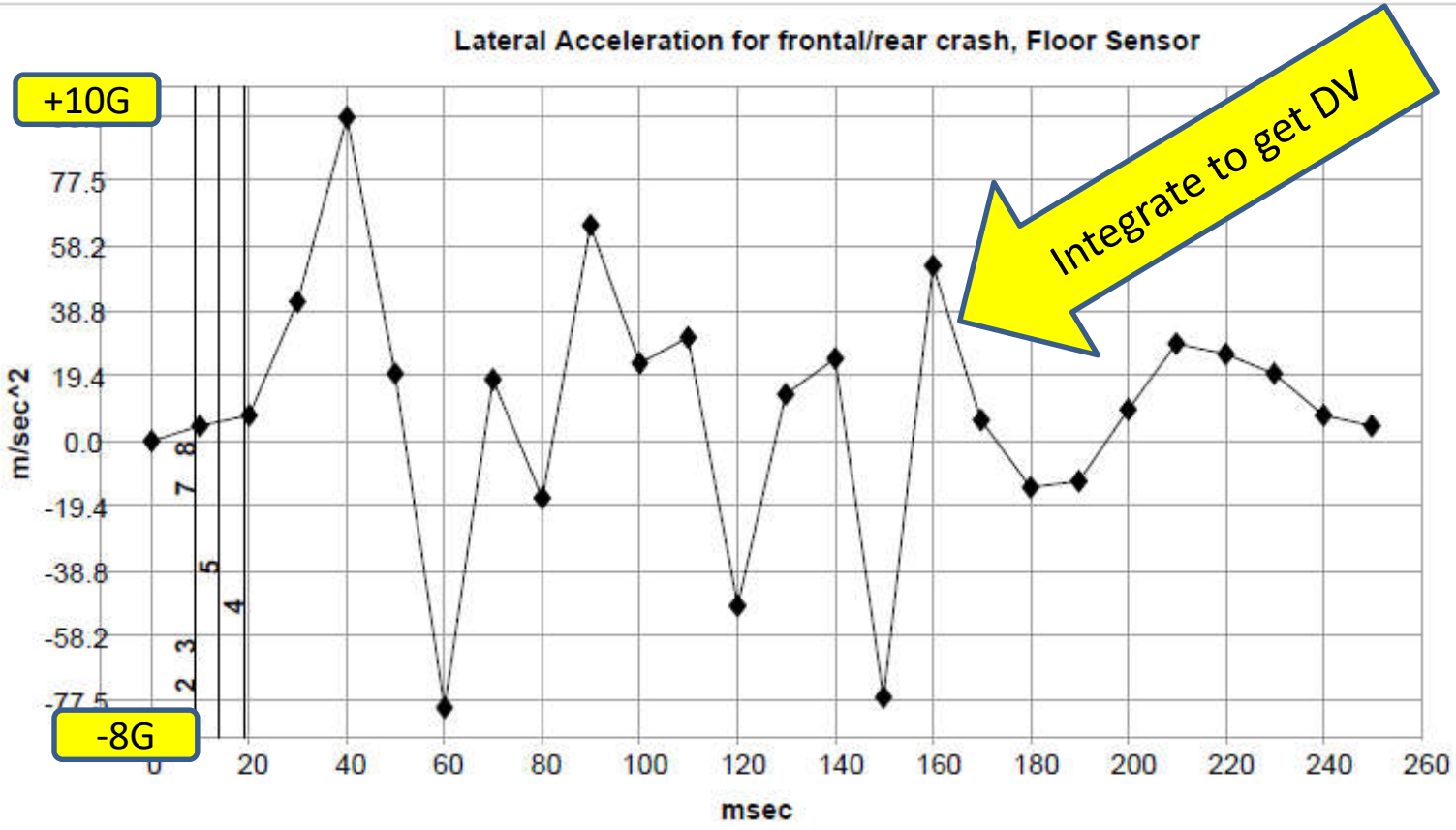
Template in Student Ref Materials

Toyota 13 EDR Lat Accel to Lat DV Template

INSTRUCTIONS
While in CDR save data is CSV format
Copy Columns A thru E from CSV file for 0-250ms
Paste into Column A thru E below. D is metric and is hidden

LONGITUDINAL/LATERAL		PASTE DATA BELOW		CALCULATIONS - IMPERIAL			CALCUALTIONS - METRIC		
Time (msec	Longitudinal Delta-V (MPH)	Lateral Acceleration for Frontal/Rear Crash, Floor Sensor (m/sec^2)	Power Supply Status	Lat DV from Floor Sensor, (MPH)	PDOF in degrees	Total DV	Long DV (KPH)	Lat DV (kph)	Total DV (kph)
0	0	0	ON	0		0.0	0.0	0.0	0.0
10	-1.2	-5.7	ON	-0.1	6.1	1.2	-1.9	-0.2	1.9
20	-3.6	-11.5	ON	-0.4	6.1	3.6	-5.8	-0.6	5.8
30	-6.7	-17.2	ON	-0.8	6.6	6.7	-10.8	-1.2	10.9
40	-9.9	-70.9	ON	-2.4	13.4	10.2	-16.0	-3.8	16.4
50	-13	-113	ON	-4.9	20.6	13.9	-21.0	-7.9	22.3
60	-15.6	-51.7	ON	-6.1	21.2	16.7	-25.1	-9.7	26.9
70	-16.5	-34.5	ON	-6.8	22.5	17.9	-26.5	-11.0	28.7
80	-17.3	-7.7	ON	-7.0	22.0	18.7	-27.9	-11.3	30.0
90	-17.5	-15.3	ON	-7.3	22.8	19.0	-28.1	-11.8	30.5
100	-17.7	-17.2	ON	-7.7	23.6	19.3	-28.4	-12.4	31.1
110	-18	0	ON	-7.7	23.2	19.6	-29.0	-12.4	31.5
120	-18.2	1.9	ON	-7.7	22.9	19.8	-29.2	-12.4	31.8
130	-18.3	0	ON	-7.7	22.8	19.8	-29.5	-12.4	31.9
140	-18.3	0	ON	-7.7	22.8	19.8	-29.5	-12.4	31.9
150	-18.3	3.8	ON	-7.6	22.5	19.8	-29.5	-12.2	31.9
$grey160	$grey0.0	$grey0.0	$greyON	#VALUE!	#VALUE!	#VALUE!	$grey0.0	#VALUE!	#VALUE!

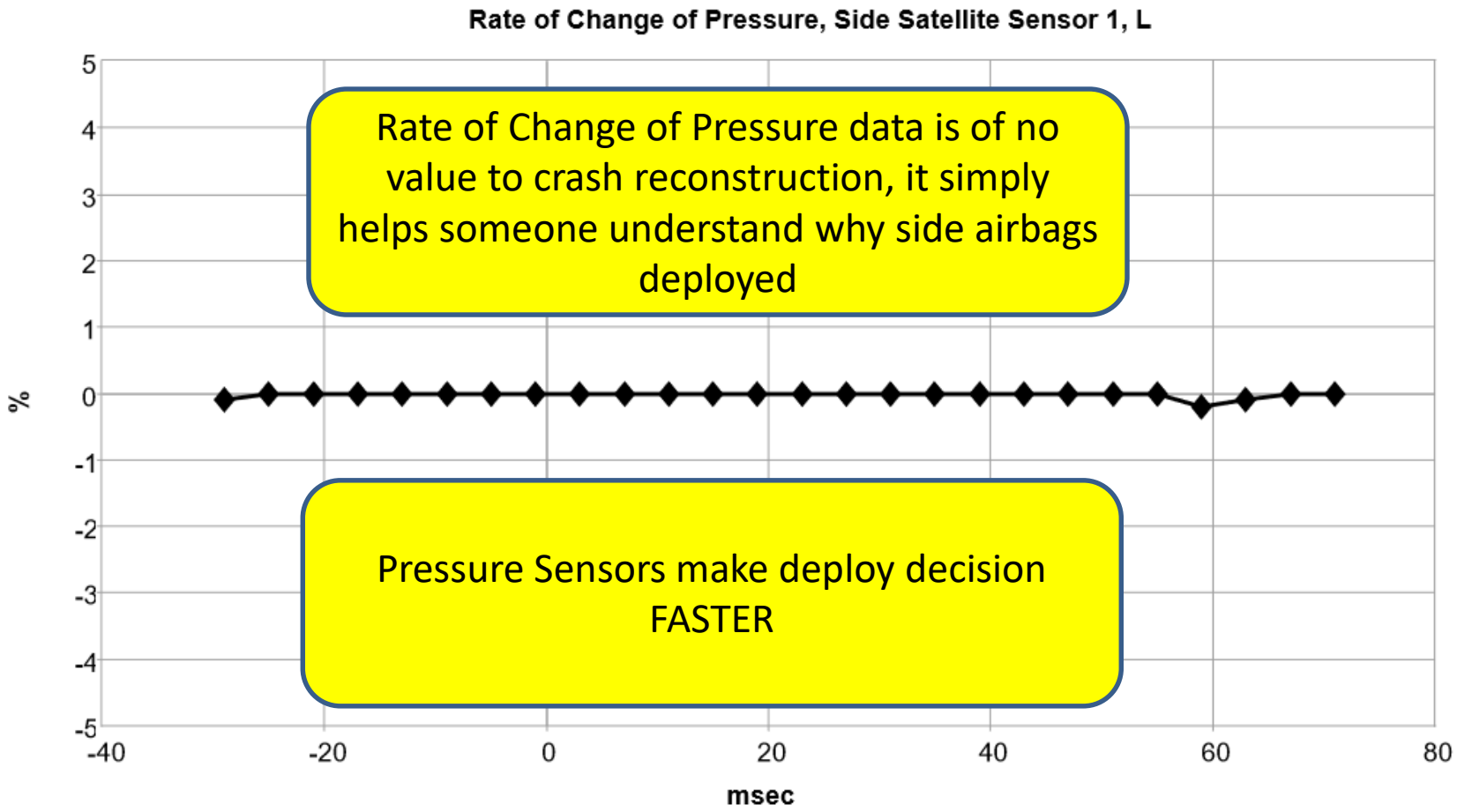
13 EDR lat accel graph from 28 mph (big) longitudinal event



Deployment Time Marker Key

1	Driver Airbag Deployment Time
2	Passenger Airbag Deployment Time
3	Driver/Passenger Pretensioner
4	Driver 2nd Stage Airbag Deployment Time
5	Passenger 2nd Stage Airbag Deployment
6	Driver/Passenger AHR
7	Driver CSA
8	Passenger CSA
9	Rear Window Airbag Deployment Time

Toyota 15 EDR
side impact door PRESSURE sensors



New “17 EDR”

- “Front/Rear” events also triggered by 5mph Y DV or side deployment
- Y Delta V from ACM added to frontal events
- More precrash registers (6 vs 2 prior)
- 4 front/rear & 4 side DV registers, 2 rollover
- Separate side events still reported, but ***do NOT have precrash associated with them***. Side events have detail of side satellite sensors (generally not used by Recons).
- Odometer data element added

17 EDR Event Record Summary
more events, detail on recording status

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash Recording Status	Diagnostic Data Recording Status	Occupant Data Recording Status	Crash Info Recording Status	Time Series Recording Status
Most Recent Event	10	Side Crash	0	N/A	Complete	N/A		Complete
1st Prior Event	9	Frontal/Rear/Side	-898	Complete		Complete	Complete	Complete
2nd Prior Event	8	ide Crash	-925	N/A			N/A	Complete
3rd Prior Event	6	Side Crash	N/A		Complete	N/A	N/A	Complete
4th Prior Event	5	Rollover	N/A	Complete	Complete	Com	Incomplete	Incomplete
5th Prior Event	4	Frontal/Rear/Side Crash	N/A	Complete	Com	Complete	Complete	Complete
6th Prior Event	3	Side Crash	N/A	N/A	Complete	N/A	N/A	Complete
7th Prior Event	2	Rollover	N/A	Complete	Complete	Complete	Complete	Complete

Pages NOT numbered like Gen3

#7 missing

Lost count due to 7 missing

No precrash for sides

17 EDR –Time in Event Record Summary "Lost" by recovery event #13, 3 key cycles after crash.

Event(s) recovered	Front/Rear/Side Events (2), Side Events (4), Rollover Events (2)

Note Severity of Each Event in the summary

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash Recording Status	Diagnostic Data Recording Status	Occupant Data Recording Status	Crash Info Recording Status	Time Series Recording Status
Most Recent Event	13	Side Crash	0	N/A	Complete	N/A	N/A	Complete
1st Prior Event	12	146° Rollover	N/A	Complete	Complete	Complete	Complete	Complete
2nd Prior Event	10	Side Crash	N/A	N/A	Complete	N/A	N/A	Complete
3rd Prior Event	9	Frontal/Rear/Side Crash	N/A	Complete	Complete	Complete	Complete	Complete
4th Prior Event	8	Side Crash	N/A	N/A	Complete	N/A	N/A	Complete
5th Prior Event	7	200° Rollover	Deployment	Complete	Complete	Complete	Complete	Complete
6th Prior Event	5	Side Crash	N/A	N/A	Complete	N/A	N/A	Complete
7th Prior Event	3	Frontal/Rear/Side Crash	-38 DVx, -20 DVy N/A Deployment	Complete	Complete	Complete	Complete	Complete

3 key cycles after 12

#11 skipped?

Lost at #13

-3 DVx, -6DVy

#6 side overwritten?

#4 side overwritten?

#3 Says #2 was not related (so we didn't miss anything important)

17 EDR –Time in Event Record Summary "Lost" by recovery event #13, 3 key cycles after crash. Compare Time Count at Event to find relative time of events

Deployment

Deployment

#3 Big hit -38 DVx

System Status at Event (7th Prior Event, TRG 3)

TRG Count (times)	3	5	7	8	9	
Event Type	Frontal/Rear/Side Crash	Side Crash	Rollover	Side Crash	Frontal/Rear/Side Crash	
Previous Crash Type	No Event	Side Crash	Side Crash	Rollover	Side Crash	
Time from Previous TRG (msec)	32767 or greater	104.5	32.5	916.0	50.0	
Time from Time Zero to TRG (msec)	1.5	ON	ON	OFF	46.5	
Event Establishment Factor	Frontal Crash	Deployment	llover CSA Deployment	None	Side Crash, Right	
TRG Establishment Factor	Frontal Crash				Side Crash, Right	
Freeze Signal	ON				OFF	
Freeze Signal Factor	loyment, Driver / Front Airbag Deployment, Passenger				None	
Recording Status, Front/Rear and Side Crash Info.	Complete				Complete	
Odometer signal (miles [km])	45,240 [72,807]	45,240 [72,807]	45,240 [72,807]	45,240 [72,807]	45,240 [72,807]	
Trip count (times)	4431	4431	4431	4431	4431	
Time count (msec)	1,711,500	1,711,600	1,713,500	1,714,300	1,714,400	
Time count input system	Normal	Normal	Normal	Normal	Normal	

Deployment Command Data (7th Prior Ever

Active Head Restraint, Time to Deploy, Driver (msec)	SNA	Complete	415.0	Complete		
Active Head Restraint, Time to Deploy, Front Passenger (msec)	SNA	0	415.0	0		
Frontal Airbag Deployment, Time to 1st Stage Deployment, Driver (	2.0	0	SNA	0		
Frontal Airbag Deployment, Time to 1st Stage Deployment, Front P	No	None	SNA			
Frontal Airbag Deployment, Time to 2nd Stage, Driver (msec)	12.0		415.0	31901), (B1906),		
Frontal Airbag Deployment, Time to 2nd Stage, Front Passenger (m	No		415.0	B1836 1821)		
Pretensioner Deployment, Time to Fire, 1st Seat, Driver (msec)	1.5					
Pretensioner Deployment, Time to Fire, 1st Seat, Passenger (msec	1.5					
Pretensioner Deployment, Time to Fire, 2nd Seat, Driver (msec)	SNA					
Pretensioner Deployment, Time to Fire, 2nd Seat, Passenger (msec	SNA		Complete			
Rear Window Airbag Deployment, Time to Deploy (msec)	SNA		0			
Side Airbag Deployment, Time to Deploy, 1st Seat, Driver (msec)	No		0			
Side Airbag Deployment, Time to Deploy, 1st Seat, Passenger (ms	140.0		(B1862), (B1837), (B1832)			
Side Airbag Deployment, Time to Deploy, 2nd Seat, Driver (msec)	SNA					
Side Airbag Deployment, Time to Deploy, 2nd Seat, Passenger (ms	SNA		10	12	13	
Side Curtain Airbag Deployment, Time to Deploy, Driver (msec)	2.0		Side Crash	Rollover	Side Crash	
Side Curtain Airbag Deployment, Time to Deploy, Passenger (msec	2.0		Frontal/Rear/Side Crash	Rollover	No Event	
			1,441.0	2,592.0	32767 or greater	
			OFF	OFF	OFF	
			None	None	None	
			45,240 [72,807]	Complete	Invalid	
			4431	4431	4434	
			1,715,800	1,718,600	115,900	
			Normal	Normal	IG	

DTCs Present at Time of Event (7th Prior Ev

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

No DTC's yet

No DTC's yet

3 DTC's

Many DTC's

DTC's take
2-3 sec to set

New **Look** to Precrash Data (same data)

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 1 of 4

Time (sec)	Vehicle Speed (MPH [km/h])	Accelerator Pedal, % Full (%)	Percentage of Engine Throttle (%)	Fuel Injection Quantity (mm^3/st)	Engine RPM (RPM)	Motor RPM (RPM)	Service Brake, ON/OFF
-5.00	29.8 [48]	49.5	24.5	Invalid	2,400	Invalid	OFF
-4.50	32.3 [52]	47.0	24.5	Invalid	2,500	Invalid	OFF
-4.00	34.2 [55]	38.0	20.0	Invalid	2,700	Invalid	OFF
-3.50	35.4 [57]	32.5	16.0	Invalid	2,800	Invalid	OFF
-3.00	36.7 [59]	26.5	13.5	Invalid	2,400	Invalid	OFF
-2.50	37.3 [60]	22.5	11.0	Invalid	2,400	Invalid	OFF
-2.00	37.3 [60]	19.0	8.5	Invalid	2,200	Invalid	OFF
-1.50	38.5 [62]	19.0	6.0	Invalid	1,900	Invalid	OFF
-1.00	37.9 [61]	18.0	5.5	Invalid	1,900	Invalid	OFF
-0.50	37.9 [61]	18.0	5.0	Invalid	1,800	Invalid	OFF
TRG(0)	36.0 [58]	100.0	26.0	Invalid	1,700	Invalid	OFF

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 2 of 4

Time (sec)	ABS Control Status	BOS Control Status	Brake Oil Pressure (Mpa)	Longitudinal Acceleration , VSC Sensor (m/s^2)	Yaw Rate (deg/s)	Steering Input (degrees)	Shift Position
-5.00	OFF	OFF	0.00	1.579	-0.98	-4.5	D
-4.50	OFF	OFF	0.00	1.795	0.00	-1.5	D
-4.00	OFF	OFF	0.00	1.292	0.49	0.0	D
-3.50	OFF	OFF	0.00	0.861	0.00	-1.5	D
-3.00	OFF	OFF	0.00	0.718	0.00	-1.5	D
-2.50	OFF	OFF	0.00	-0.072	0.00	-1.5	D
-2.00	OFF	OFF	0.00	0.000	0.00	-1.5	D
-1.50	OFF	OFF	0.00	0.000	-0.49	-1.5	D
-1.00	OFF	OFF	0.00	0.287	0.00	-1.5	D
-0.50	OFF	OFF	0.00	-0.144	0.00	-1.5	D
TRG(0)	OFF	OFF	0.48	-1.507	61.00	-19.5	D

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 3 of 4

Time (sec)	Sequential Shift Range	Cruise Control Status	VSC Control Status	READY Signal	Drive Mode, Power Train	Drive Mode, Snow	Drive Mode, EV
-5.00	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid
-4.50	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid

17 EDR time data & ped events

System Status at Pedestrian Event (1st Prior Event, TRG 1)

TRG Count (times)	1
TRG Count not for Pedestrian (times)	SNA
Ignition Cycle, Crash (times)	353
Odometer signal (miles [km])	1,195 [1,923]
Trip count (times)	392
Time count (msec)	1,283,900
Time count input system	Normal
Airbag Warning Lamp, On/Off	OFF
Pedestrian Warning Lamp, On/Off	OFF
Time from Previous TRG (msec)	32767 or greater
Time from Pre-Crash to TRG (msec)	300
Time between TRG not for Pedestrian and TRG for Pedestrian (msec)	SNA
TRG Establishment not for Pedestrian, within 5 sec.	ON
Freeze Signal	ON
Recording Status, Pre-Crash	Complete
Recording Status, Crash Info.	Complete
Recording Status, EDR Correlation Info.	Complete

Not present in side events

Similar to Ignition Cycles but slightly higher

Ignition Time on to event

Deployment Command Data (1st Prior Event, TRG 1)

Front Pop-Up Hood, Time to Deploy (msec)	3.0
Rear Pop-Up Hood, Time to Deploy (msec)	3.0
Airbag for Pedestrian Deployment, Time to Deploy (msec)	SNA

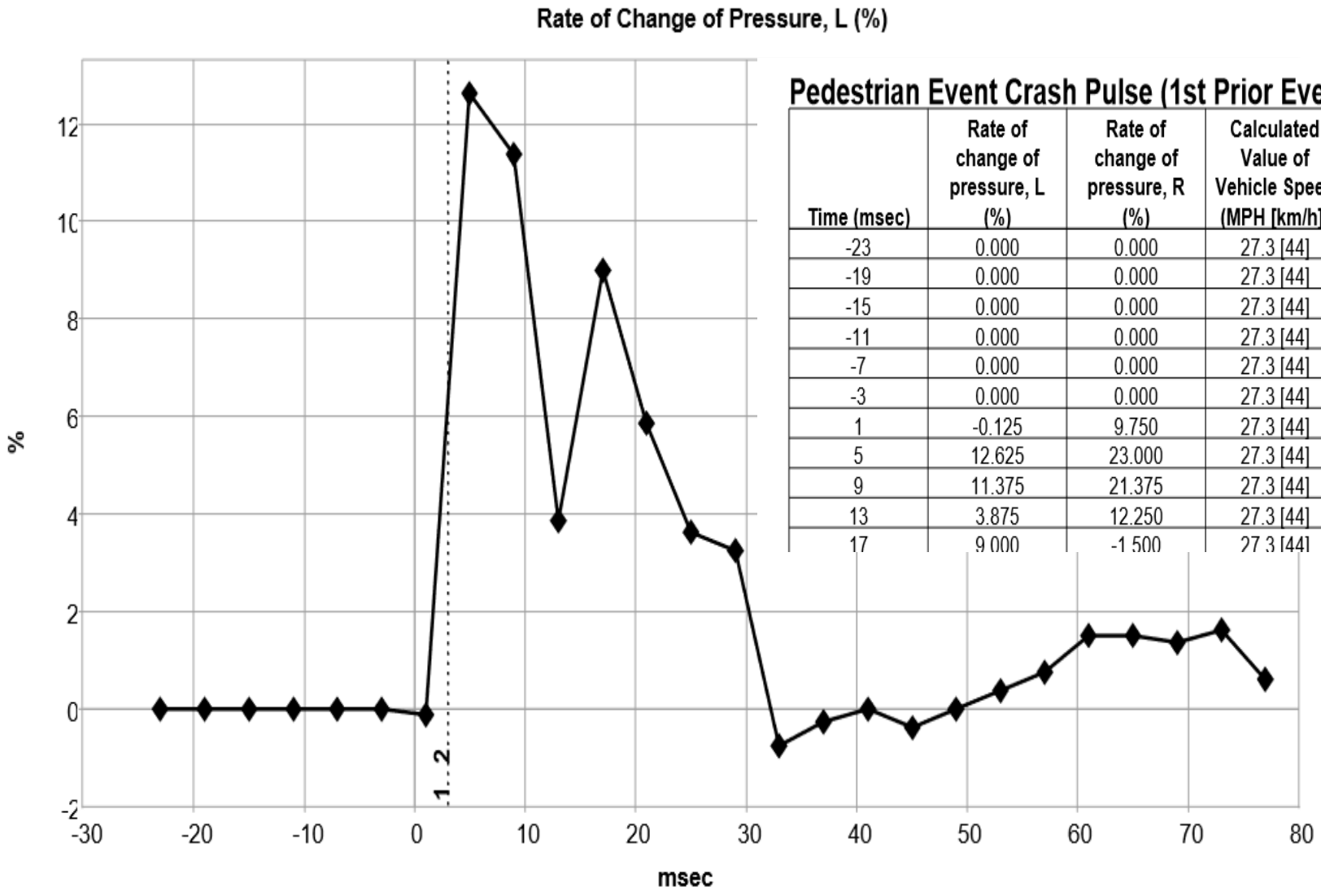
DTCs Present at Time of Event (1st Prior Event, TRG 1)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Warning ON Time Since DTC was Set, for Pedestrian (min)	0
Diagnostic Trouble Codes	None

Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)

Recording Status, Time Series Data	Complete
Time from TRG to Next Sample (msec)	1

Data from L/R bumper pressure sensors



Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)

Time (msec)	Rate of change of pressure, L (%)	Rate of change of pressure, R (%)	Calculated Value of Vehicle Speed (MPH [km/h])
-23	0.000	0.000	27.3 [44]
-19	0.000	0.000	27.3 [44]
-15	0.000	0.000	27.3 [44]
-11	0.000	0.000	27.3 [44]
-7	0.000	0.000	27.3 [44]
-3	0.000	0.000	27.3 [44]
1	-0.125	9.750	27.3 [44]
5	12.625	23.000	27.3 [44]
9	11.375	21.375	27.3 [44]
13	3.875	12.250	27.3 [44]
17	9.000	-1.500	27.3 [44]

New VCH Freeze Frame Data

- New feature in 2013+ Rav 4 and 2014+ Highlander & Lexus IS, 2015+ Lexus NX & RC, "**V**ehicle **C**ontrol **H**istory"
- Triggers EDR-style recording if accel pedal is rapidly depressed or other conditions that might occur during an alleged sudden or unintended acceleration or lane excursion. 26 triggers including crash avoidance, ESC.
- Accessible only with Toyota Techstream service tool – not part of Bosch CDR system. Annual s/w subscription or 2 day subs available.
- CDR900 or Nexiq works as pass thru device same as Toyota Mongoose hardware
- Data exports as CSV to Microsoft excel
- Space in memory for MANY events – can show a history of erratic driver behavior
- Details in SAE 2016-01-1495 Appendix D. New **2019-01-0632** "**Reconstructing Vehicle Dynamics from On-Board Event Data**" **discovered VCS has data 6.6 times per second in ABS events vs 2 times in hard accel events).**
- **Photos stored by pre-collision braking system may be accessed with Techstream system!**

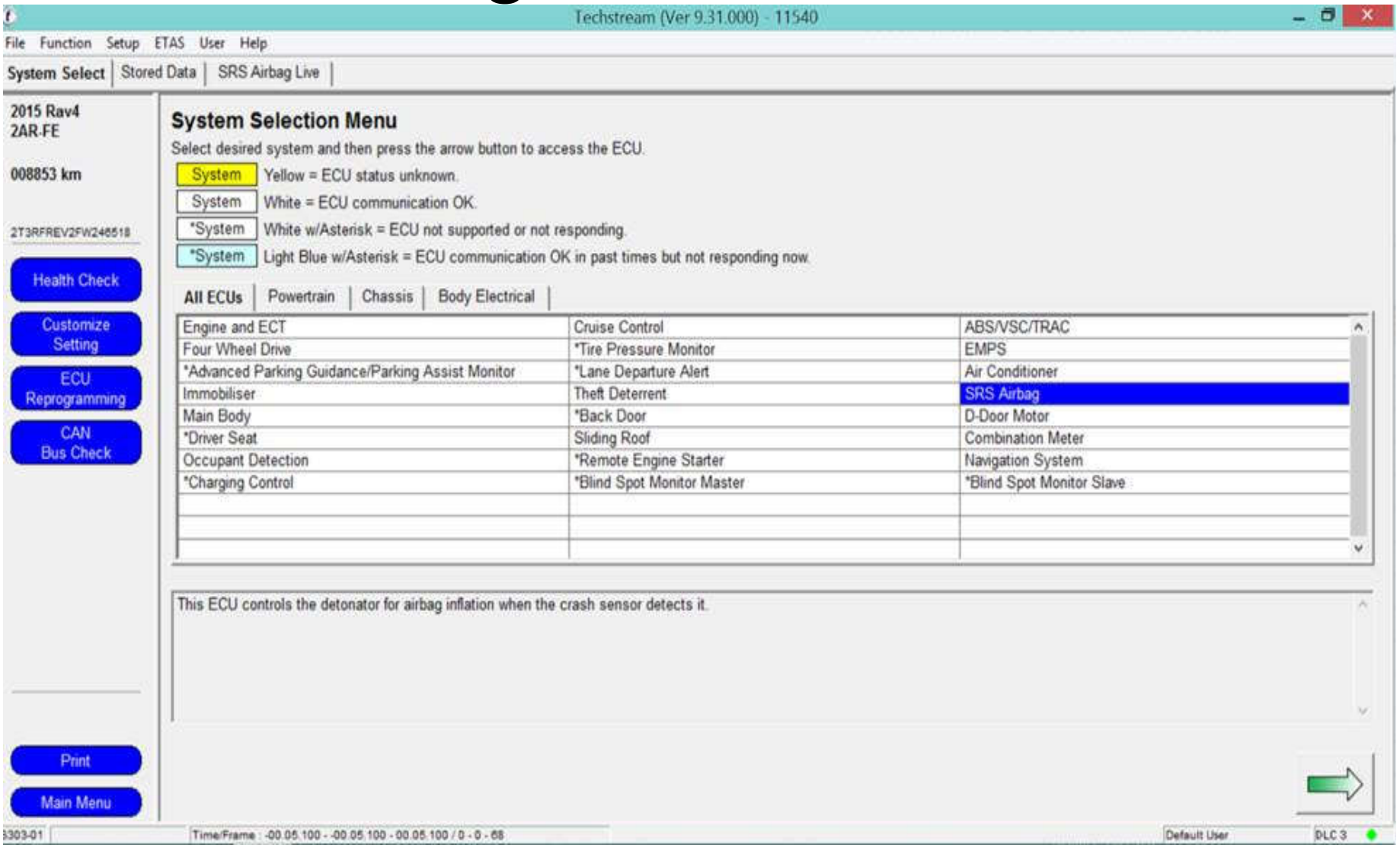
131-slide presentation by Brad Muir has details on accessing the VCH, and new crash avoidance data



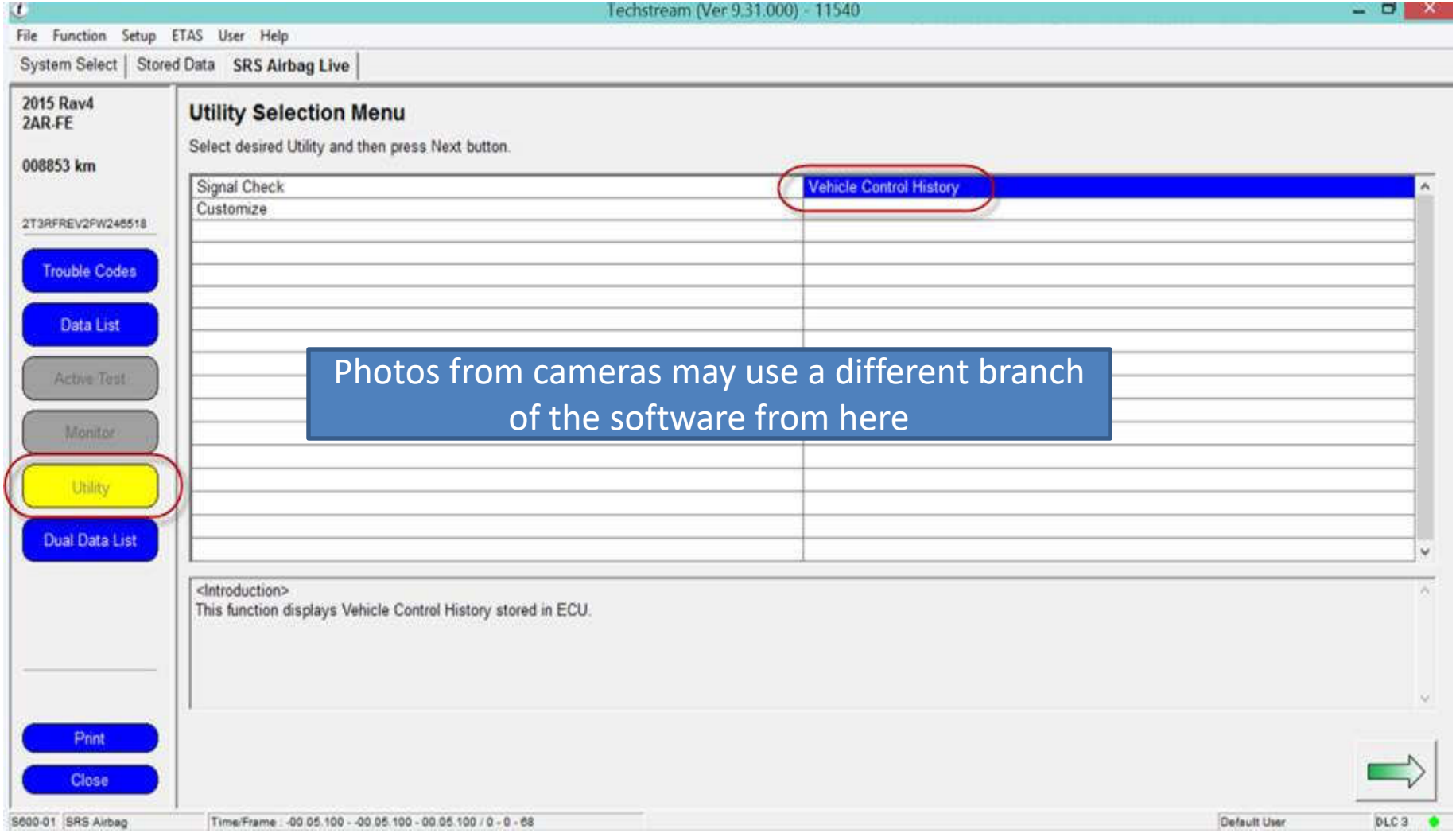
TOYOTA/LEXUS Vehicles with Vehicle Control History Data

		2013	2014	2015	2016	2017	2018	2019	2020
LEXUS	CT200h								
	ES								
	GS								
	GX								
	IS		250 and 350						
	LS								
	LX								
	NX								
	RC								
	RX								
TOYOTA	Avalon								
	Camry								
	Corolla								
	Prius			C and V	and Prius	and Prime			
	Highlander								
	Land Cruiser								
	4Runner								
	Tacoma								
	Tundra								
	Sequoia								
	RAV4								
	Sienna								
	Yaris								
SCION	iM								

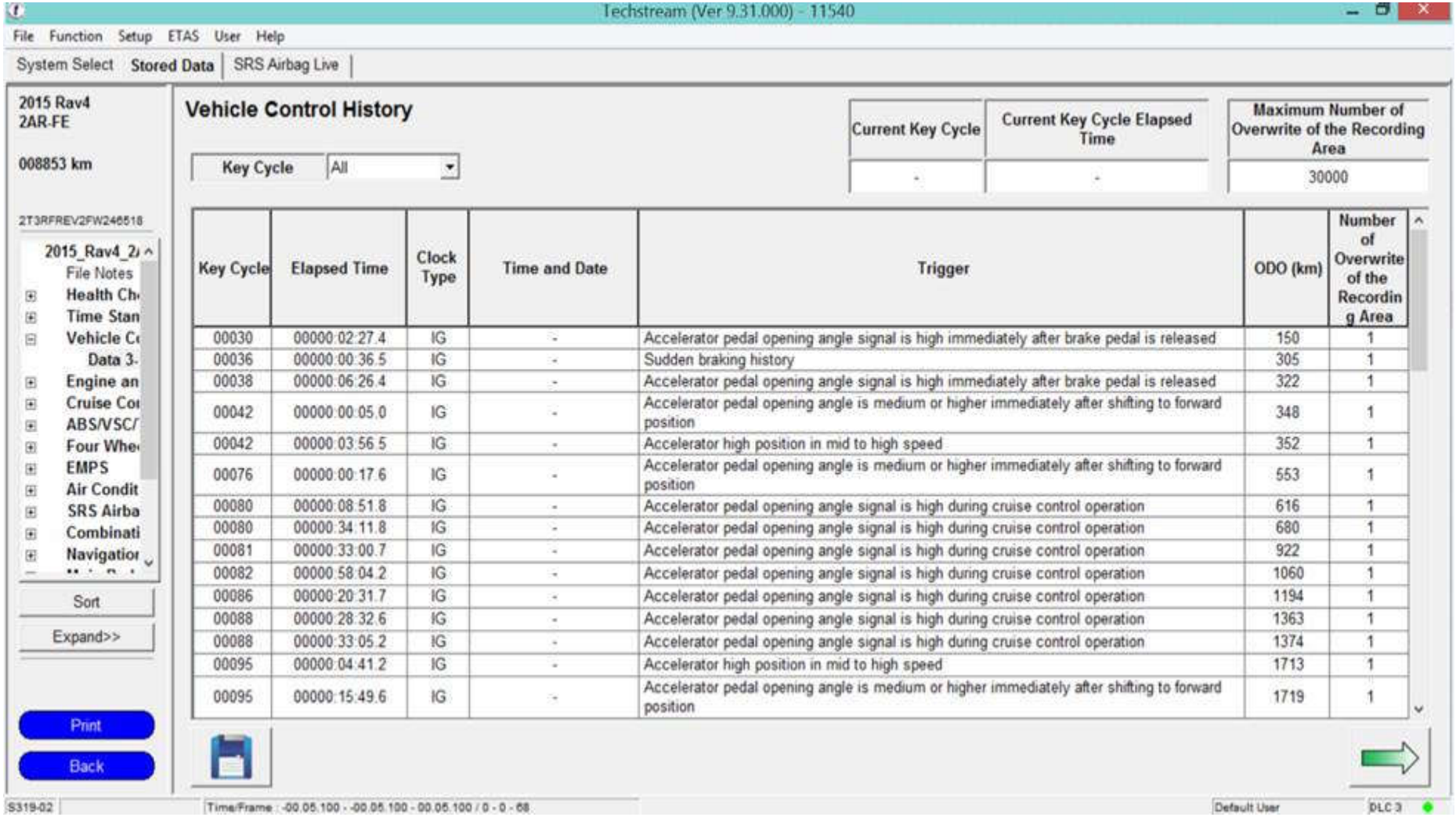
Getting to Vehicle Control History Using Techstream Tool



Using Techstream to Access VCH data

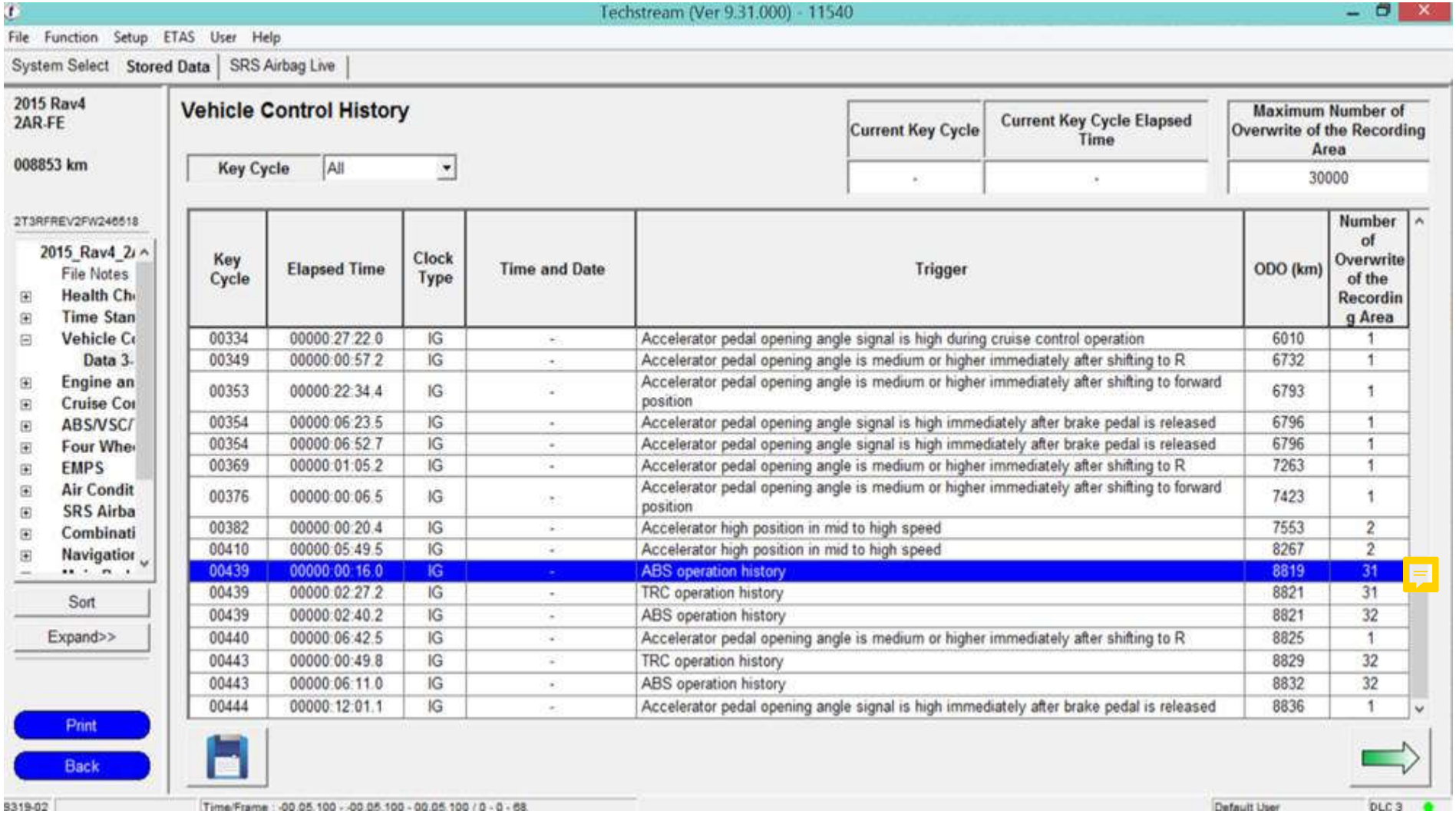


VCH List of Recorded Events



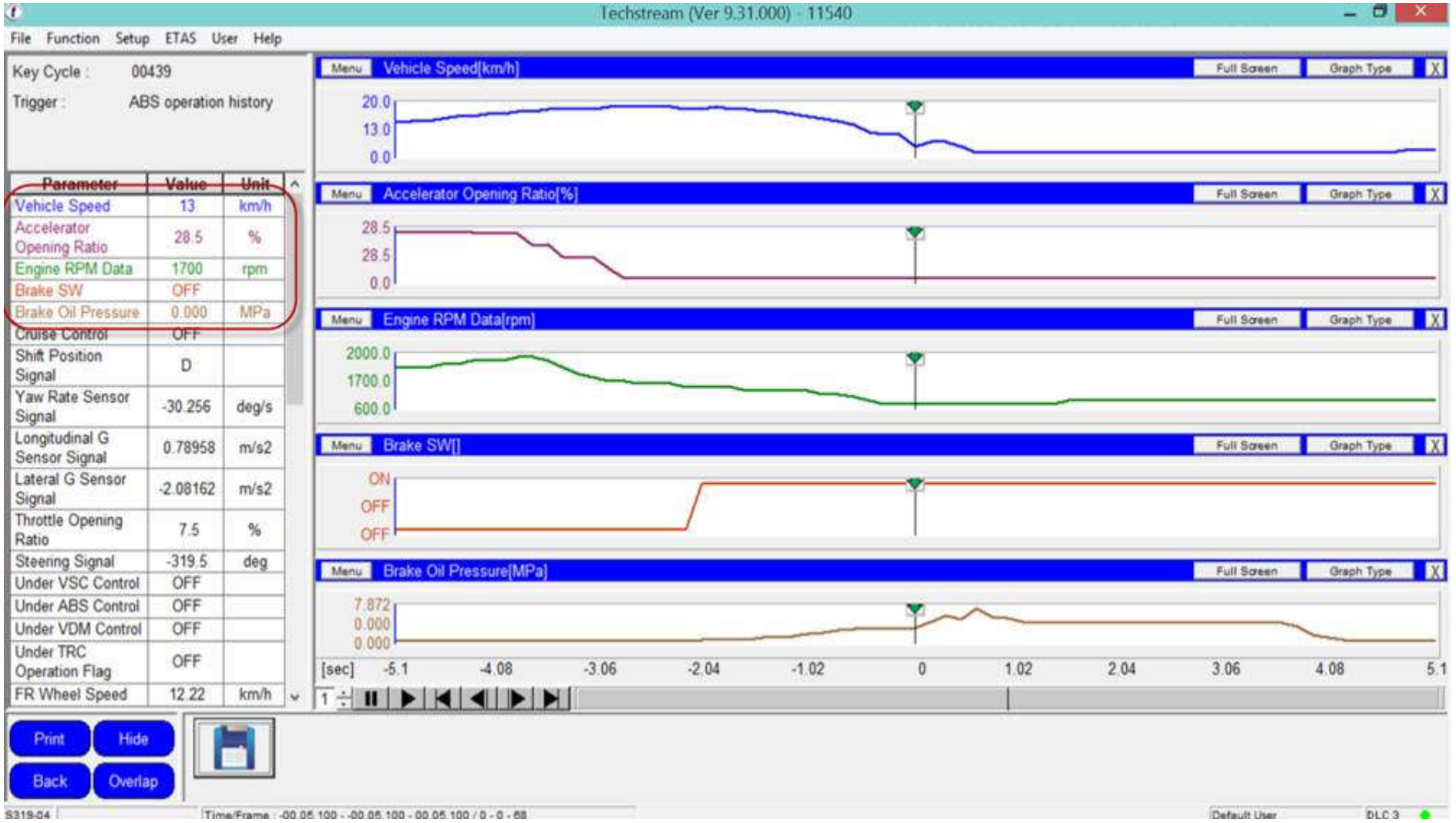
Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	ODO (km)	Number of Overwrite of the Recording Area
00030	00000:02:27.4	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released	150	1
00036	00000:00:36.5	IG	-	Sudden braking history	305	1
00038	00000:06:26.4	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released	322	1
00042	00000:00:05.0	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	348	1
00042	00000:03:56.5	IG	-	Accelerator high position in mid to high speed	352	1
00076	00000:00:17.6	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	553	1
00080	00000:08:51.8	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	616	1
00080	00000:34:11.8	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	680	1
00081	00000:33:00.7	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	922	1
00082	00000:58:04.2	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	1060	1
00086	00000:20:31.7	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	1194	1
00088	00000:28:32.6	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	1363	1
00088	00000:33:05.2	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	1374	1
00095	00000:04:41.2	IG	-	Accelerator high position in mid to high speed	1713	1
00095	00000:15:49.6	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	1719	1

Highlight Entry To Get Detail



Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	ODO (km)	Number of Overwrite of the Recording Area
00334	00000:27:22.0	IG	-	Accelerator pedal opening angle signal is high during cruise control operation	6010	1
00349	00000:00:57.2	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to R	6732	1
00353	00000:22:34.4	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	6793	1
00354	00000:06:23.5	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released	6796	1
00354	00000:06:52.7	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released	6796	1
00369	00000:01:05.2	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to R	7263	1
00376	00000:00:06.5	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position	7423	1
00382	00000:00:20.4	IG	-	Accelerator high position in mid to high speed	7553	2
00410	00000:05:49.5	IG	-	Accelerator high position in mid to high speed	8267	2
00439	00000:00:16.0	IG	-	ABS operation history	8819	31
00439	00000:02:27.2	IG	-	TRC operation history	8821	31
00439	00000:02:40.2	IG	-	ABS operation history	8821	32
00440	00000:06:42.5	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to R	8825	1
00443	00000:00:49.8	IG	-	TRC operation history	8829	32
00443	00000:06:11.0	IG	-	ABS operation history	8832	32
00444	00000:12:01.1	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released	8836	1

Detailed VCH Data Per Event



Toyota EDR Accuracy

- Toyota stated up front in 2009 that its first prototype readout tool was not validated –but that didn't mean it wasn't right.
- Because early Toyota EDR's (like many other manufacturers) that lost power at impact may not make a new recording at impact, there were examples of when the Toyota EDR's did not match the accident facts.
- Some claimed the Toyota EDR was unreliable - based on a lack of understanding of the not-yet-written data limitations.

Specific Toyota Validation

- Niehoff 2005: 3 barrier crashes: Delta V ranged -6.7% to +1.1%, only vehicle with speed data was -0.9% of reference instrument.
- Gabler 2008: 11 frontal/4 side barrier crashes: Delta V data ranged from -22% to +11%, avg -1.6%. Only one vehicle had speed data, it reported 35 mph in a 33.8 mph test or 3% high. Belt status was reported correctly in all 15 tests
- NHTSA 2011: 14 staged events, “No errors in measurement and storage or pre crash data” (within known accuracy and timing limitations). Vehicle speed within 1.5 mph, throttle voltage within 2.5%.

Additional Toyota Validation

- Ishikawa & Takubo 2009: 14 J-NCAP crashes EDR Delta V is from -14% to +0.5% of reference instrumentation, averaging -7.3%. Speed data within 4%, avg. 2%
- Ruth 2012: SAE 2012-01-0999 -Speed data within 3%
- Brown 2012: SAE 2012-01-0996 & 0998 (Actual speed -2.7%)
- Wilkinson 2013: SAE 2013-01-1268 (Gen 1 Delta V 1mph low in low speed frontal collisions, 1mph high in rear collisions)
- Webster 2014: SAE 2014-01-0505 **Accuracy of Recorded Driver Inputs in Toyota Part 563 EDR** (Only addressed accuracy of brake and accelerator pedal position).
- SAE 2016-01-1494, 2017-01-1438 continue Wilkinson 2013 work into other generations. Gen 1-2 ACM Delta V may be under reported slightly in frontal collisions, and over reported slightly in rear collisions

Additional Toyota Validation

2019-01-0633 Performance of Event Data Recorders Found in Toyota Airbag Control Modules in High Severity Frontal Oblique Offset Crash Tests by William Bortles, Kineticorp et al (finds Delta V is understated due to offset).

2019-01-0635 Laboratory Experiments Using a 2007 Toyota Auris Event Data Recorder and Additional Data

from CAN Bus by Miguel Tabone et al, University of Malta (took electrical system out of car into lab, injected ABS wheel speed sensor signals to simulate moving vehicle as events were set by hammer blows)

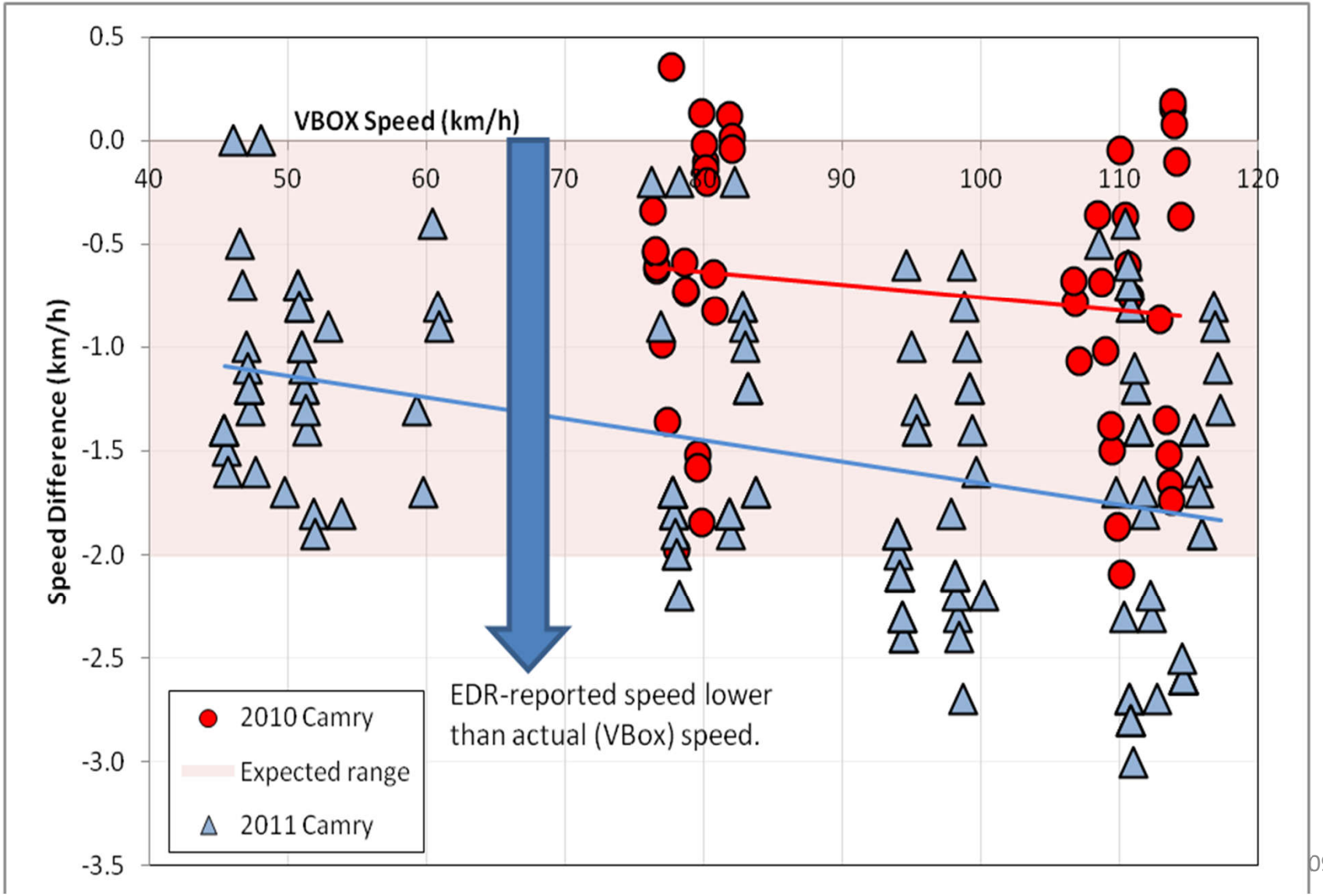
2019-01-0632 Reconstructing Vehicle Dynamics from On-Board Event Data by Brandon Tsuge et al, MEA Forensic (read out Vehicle Control History, discovered VCS has data 6.6 times per second in ABS events vs 2 times in hard accel events)

CONCLUSION

When the recording is from the event of interest, and data limitations are understood, there is no reason to believe speed data is any less accurate than any other manufacturer's EDR based on 42 test events. Delta V may under report slightly in Gen 1 and Gen 2 ACM's in frontal collisions and over reported in rear collisions, but in barrier tests are still within the generally accepted +/-10%. Other than the 00 EDR seat belt reporting at time of readout and not event (noted in the data limitations), there are no known issues with other data elements at this time.

Toyota Speed Data Accuracy Validated

SAE 2012-01-0999 2010/11 Camry VBOX vs EDR



The Accuracy and Sensitivity of 2005 to 2008 Toyota Corolla Event Data Recorders in Low-Speed Collisions
SAE 2013-01-1268

- Induced a series of low speed events on vehicles and on a sled
- Verified 2G wakeup threshold thru testing
- Determined “offset” in Toyota accelerometers causing them to read 1.3 to 2.6 km/h low in frontal collisions and 0.6 to 2.2 km/h high in rear collisions.

Toyota EDR Summary

- Toyota began installing EDR's in 2001MY; many Gen 1 and some Gen 2 have Delta V data only, others have DV and precrash data. Gen 1 & 2 data is similar to GM but has added precrash data point sampled at TRG.
- Unique Gen 1 &2 issues (75/78 mph max speed, accel pedal in volts, & last data point up to 0.5 seconds old) require special attention.
- Toyota EDR data is as ***reliable*** as other manufacturers within the data limitations.
- "12EDR" precrash has many additional data elements helpful to crash recon. But has no ACM Y Delta V in frontal events.
- "13EDR" adds synchronized Y accel data.
- New Vehicle Control History more frequent data, pics!!!
- Newest 17 EDR has synchronized X/Y Delta V